Business Analytics Toolkit Portfolio Ch 03–05

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Course: BUSI 6532 - Business Analytics & Data Visualization

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Chapter 1 SAP Analytics Cloud

1.1 Introduction of SAP Analytics Cloud

SAP Analytics Cloud (SAC) is a cloud-based analytics tool that helps organizations analyze data and present it in a clear visual format. Instead of working through large tables of numbers, users can create charts, dashboards, and interactive reports that make patterns and trends easier to understand. One of the main advantages of SAP Analytics Cloud is that it brings data visualization, analysis, and business planning together in one platform. It can connect to different data sources and transform raw datasets into visual stories that highlight important insights. Because of this, SAC is widely used in businesses for performance tracking, reporting, and supporting data-driven decisions.

1.2 Dataset and Research Questions

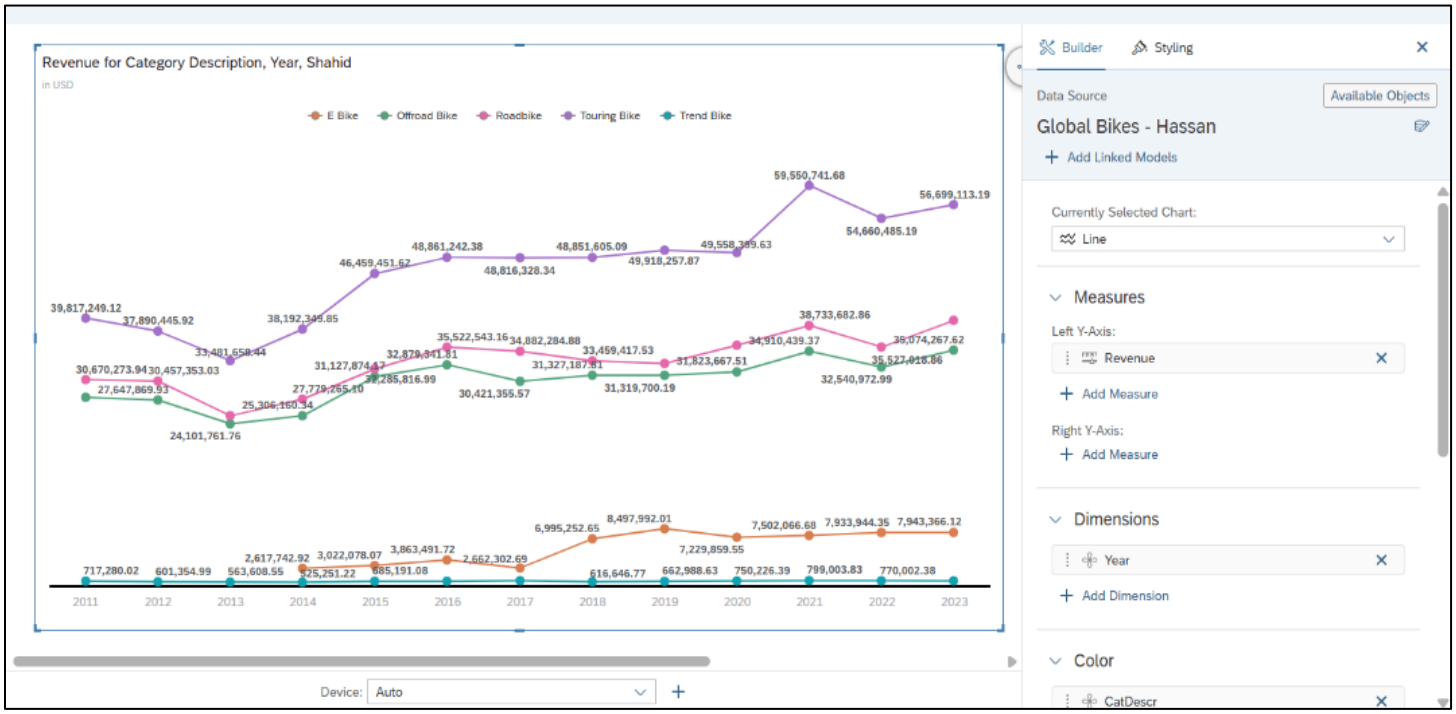
This section presents the analysis conducted using SAP Analytics Cloud based on the Global Bike Inc. wholesale dataset. The dataset consists of approximately 29,000 transactional records, covering multiple years of wholesale sales activity from 2011 to 2023. Key data fields used in the analysis include Calendar Year and Month, Customer Description, Country, Region, Sales Quantity, Revenue, Net Sales, Cost of Goods, and Discount. Each research question is addressed using a dedicated pivot table and supported by a corresponding visualization, allowing patterns and comparisons to be identified clearly below:

- Which product category (E-Bike, Offroad, Road, Touring, or Trend) has seen the most significant growth in revenue share between 2011 and 2023?
- Within the United States, which specific locations (e.g., Silicon Valley Bikes vs. Peachtree Bikes) are the top three contributors to sales volume?
- Which customers maintain the highest Gross Margin Ratio, and is there a negative correlation between high sales volume and high margin percentage?
- Based on the revenue trends from 2011 to 2023, what is the projected revenue forecast for the first half of 2024, and what is the degree of uncertainty (confidence bound) for those months?
- How do high-volume, low-profit products impact the Sales Organisation efficiency?
- What are the primary drivers of revenue leakage across Global Bike's operations, and how do discounting strategies impact the profit efficiency of its largest customers?

1.3 Deriving Business Insights and Findings

1.3.1. Which product category (E-Bike, Offroad, Road, Touring, or Trend) has seen the most significant growth in revenue share between 2011 and 2023?

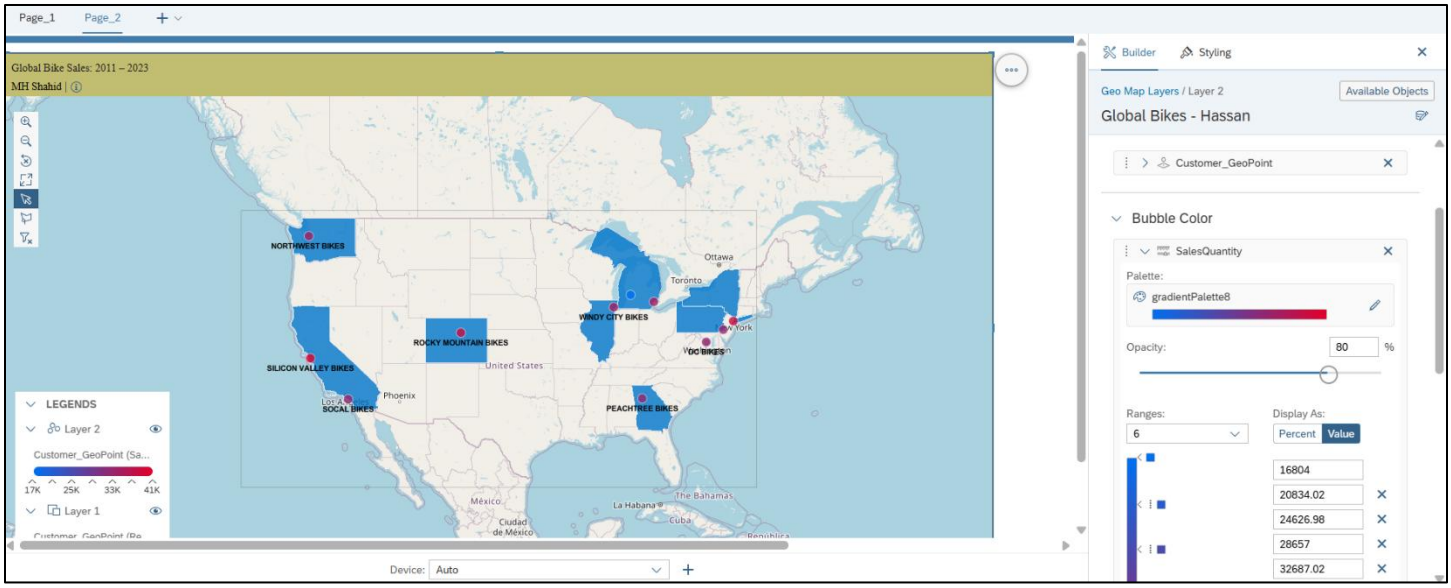
To capture the long-term trends of our product lineup, I built a line chart within the SAP Analytics Cloud. I placed "Year" on the horizontal X-axis and "Revenue" on the vertical Y-axis to track our financial performance over the 2011–2023 period. The "secret sauce" that really brought the data to life was dragging the "Category Description" into the "Color" slot; this instantly separated the various bike types into distinct, color-coded lines that made comparisons effortless as shown in Fig.1. Without a doubt, the Touring category has seen the most aggressive growth. Back in 2011, it was competing with Road bikes, but it skyrocketed to nearly $60M by 2021. Touring Bikes overtook Road Bikes in total revenue around 2012. Over the last three years, E-Bikes have maintained a much higher, steadier revenue stream. Trend Bikes, while interesting, remain a very small slice of the pie; their revenue line is nearly flat and sits at the bottom of the chart.



(Fig.1)

1.3.2. Within the United States, which specific locations (e.g., Silicon Valley Bikes vs. Peachtree Bikes) are the top three contributors to sales volume?

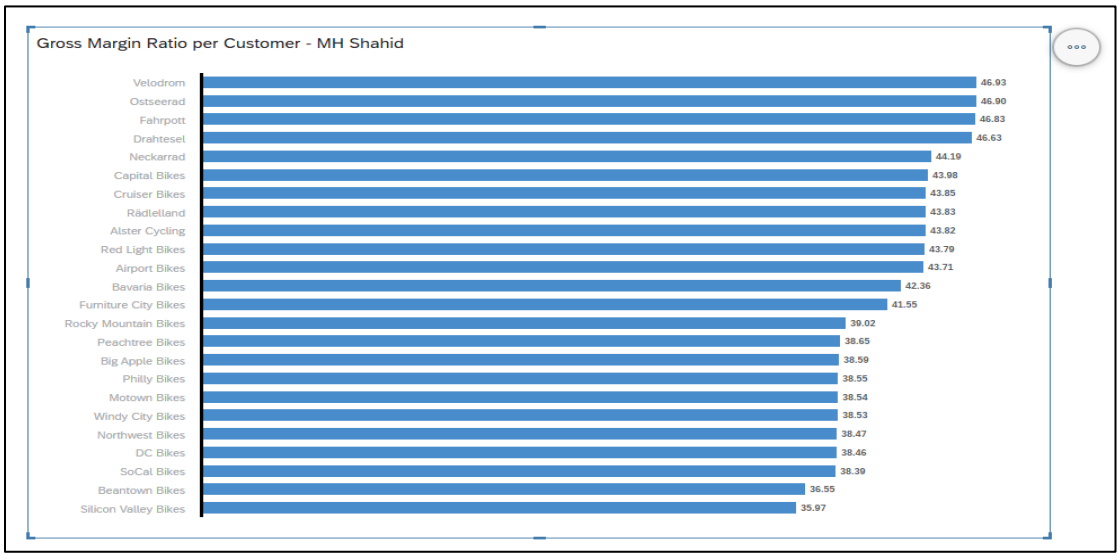
I used the Geo Map widget in which I selected the Choropleth layer, linked it to the "Customer_GeoPoint" dimension, and set the "Color" to "SalesQuantity". I then used the Hierarchy button to drill down from Country to Region. From the Fig.2, The top three contributors in the US are Peachtree Bikes, Silicon Valley Bikes, and Rocky Mountain Bikes.



(Fig.2)

1.3.3. Which customers maintain the highest Gross Margin Ratio, and is there a negative correlation between high sales volume and high margin percentage?

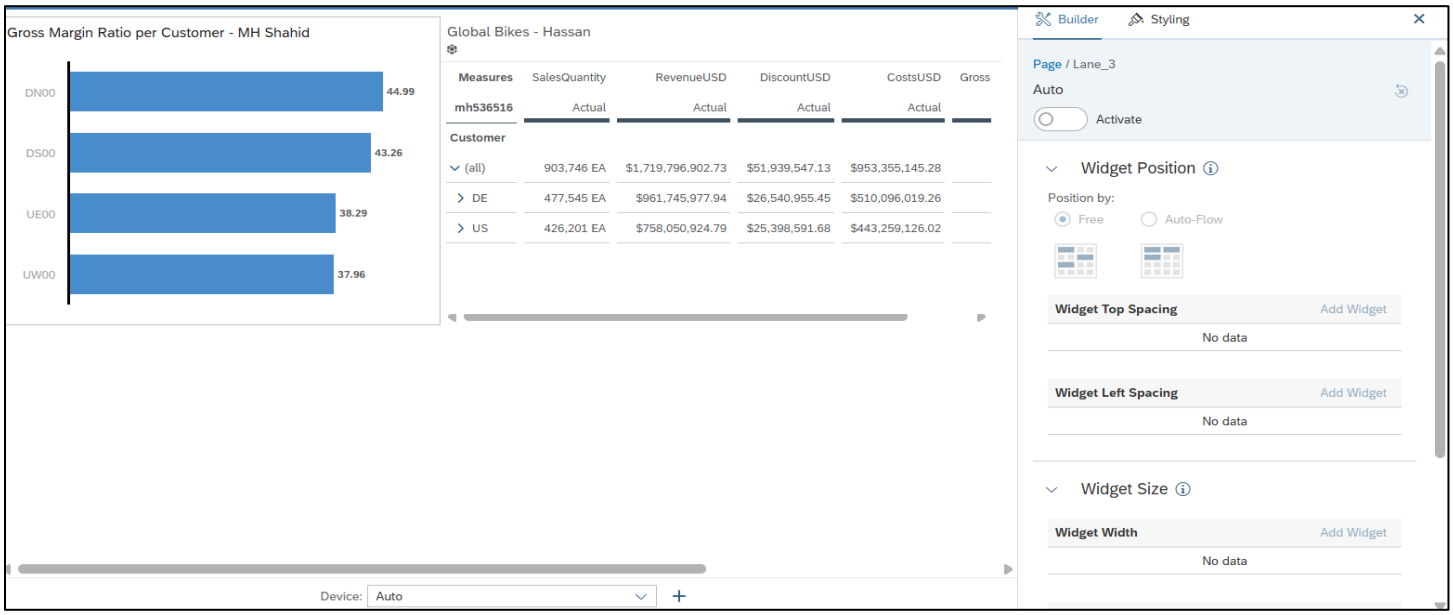
I built a combination of a Bar Chart and a Comparison Table in SAC. The main thing here was creating a calculated measure for the gross margin ratio. I created a calculated measure using the formula: Gross Margin Ratio = (Revenue – Cost)/(Revenue) and then sorted the chart by this new measure to see who was actually the most profitable. I then sorted the bar chart by this ratio to rank our customers from most to least efficient. To really see the "why" behind the numbers, I added a table that cross-referenced Discounts (USD) against the final margin, which allowed me to spot exactly where the profits were being "given away." When we look at the data, there's a slight negative correlation appearing. For example, some high-volume shops in the US are currently sitting at a lower margin of around 36% to 39%. However, one name absolutely dominates the efficiency ranking: Velodrom as shown in Fig.3.



(Fig.3)

Despite being a massive, high-volume customer, they managed to maintain a "best-in-class" Gross Margin Ratio of approximately 47%. When we shift our gaze to regional efficiency, the (DN00) is the clear winner. They operate at a cost-to-revenue efficiency that is consistently 2 to 3 percentage points higher than their (UW00) counterparts. The data in Fig.4

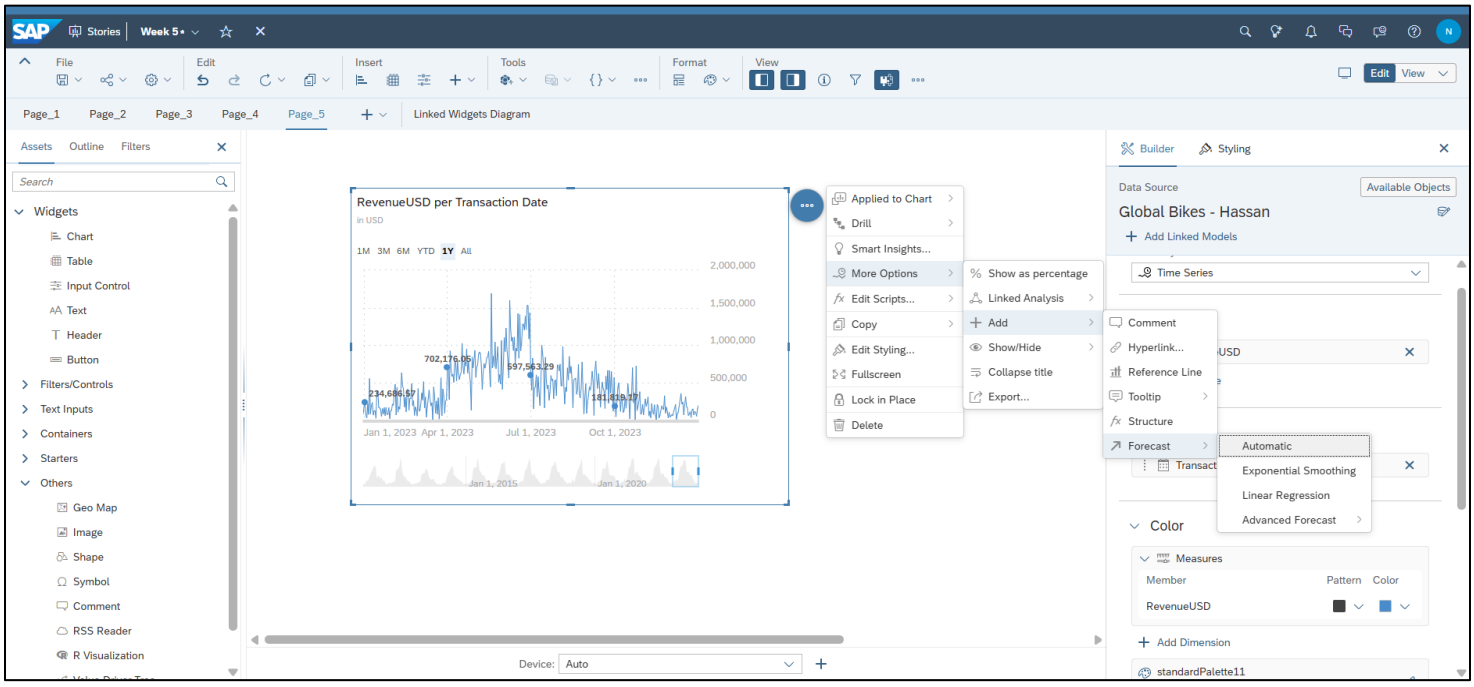
also reveals a harsh reality about pricing strategy: discounts are true "margin killers." Based on the top 10 customers, we can see that every $10,000 increase in discounts roughly correlates to a 1.5% drop in the final Gross Margin.



(Fig.4)

1.3.4. Based on the revenue trends from 2011 to 2023, what is the projected revenue forecast for the first half of 2024, and what is the degree of uncertainty (confidence bound) for those months?

I dragged the Time Series chart widget onto the canvas and then set the time dimension to "Transaction Date" and the measure to "RevenueUSD." To get the forecast, I clicked the three dot icon in the and selected "More Actions" menu, then added "Automatic Forecast" to give forecasting as shown in Fig.5 below.



(Fig.5)

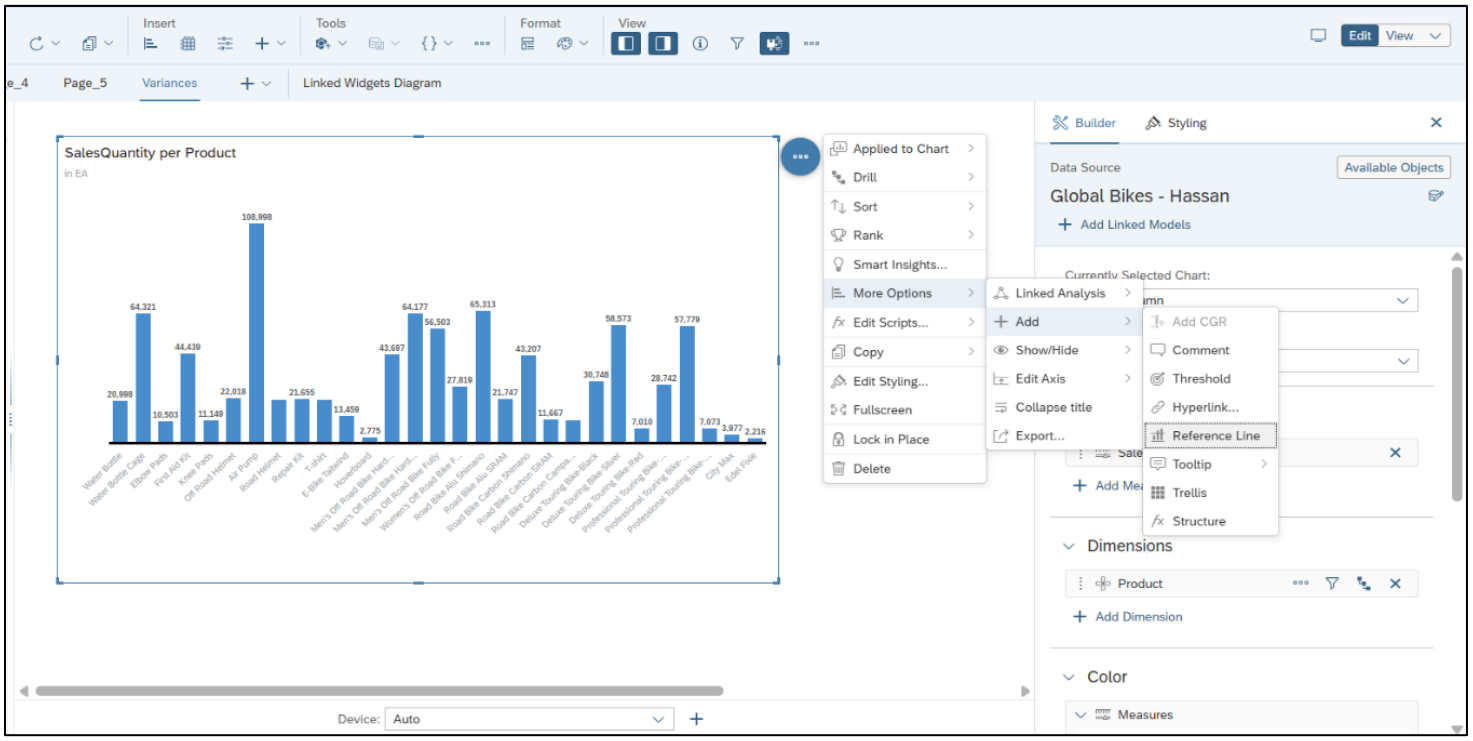
Based on the upward trend since 2011, the forecast for the first half of 2024 looks optimistic but volatile. For March 2024, the expected revenue is roughly $398,248. The degree of uncertainty (that shaded "confidence bound" area) is actually quite wide ranging from about $250k on the low end to $1,540k on the high end as shown in Fig.6 below.



(Fig.6)

1.3.5. How do high-volume and low-profit products impact the Sales Organisation efficiency?

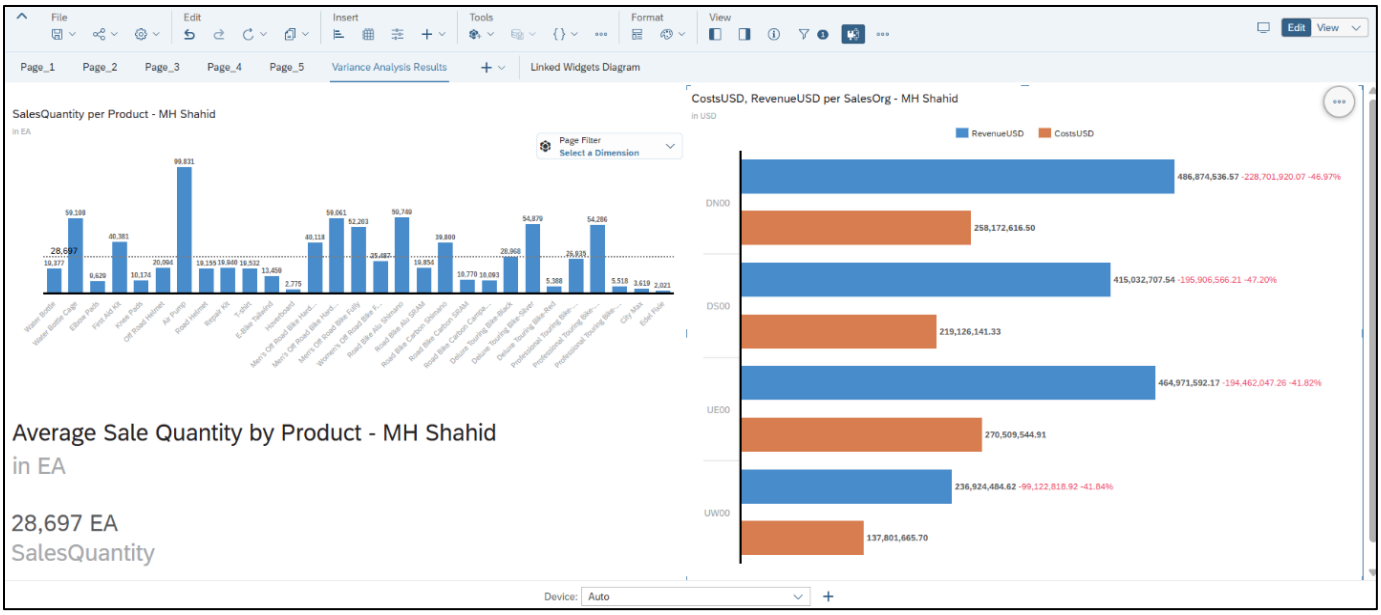
To investigate the relationship between what we sell and how much it costs us to sell it, I constructed a multi-lane dashboard. First, I created a Column Chart to track SalesQuantity per Product, adding a Reference Line to represent the average sales volume. This allowed me to immediately identify "logistics-heavy" outliers as shown in fig.8 below.



(Fig.7)

Next, I added a second lane with a Variance Analysis chart. I mapped RevenueUSD and CostsUSD across the Sales Organizations (SalesOrg) and configured a variance comparison where I compared CostsUSD with RevenueUSD. The results reveal a significant operational burden. As shown in Fig.8, products like the Air Pump (99,831 units) and Water

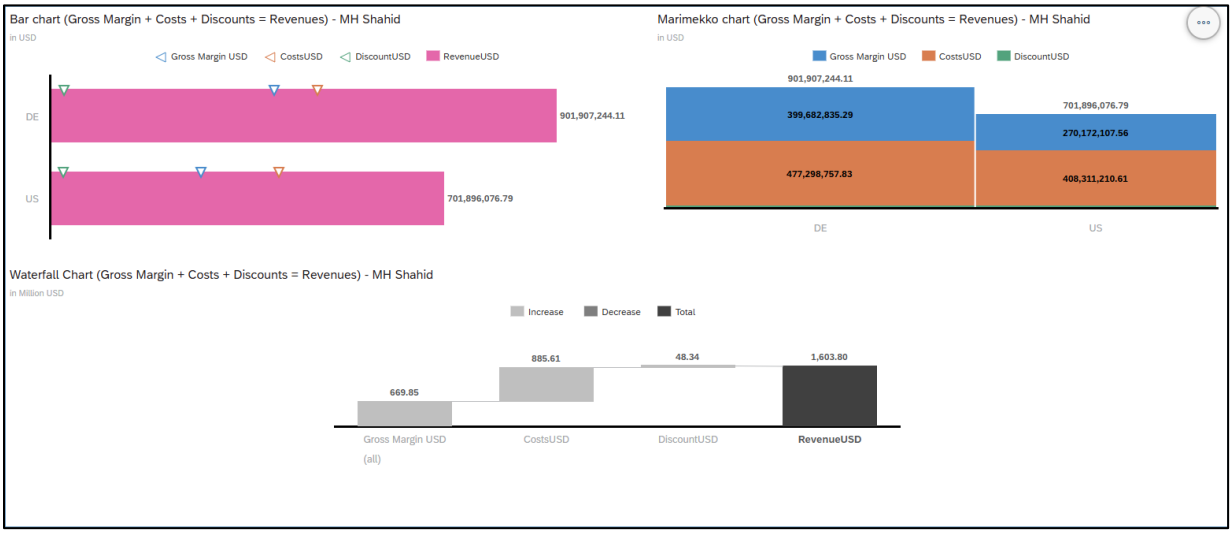
Bottle Cage (59,108 units) move in volumes nearly triple the corporate average of 28,697 units. When we pivot to the Variance Analysis by region, we see how this volume affects regional efficiency. DN00 generates the highest revenue at $486.8M, it carries a heavy cost variance of -46.97%. DS00 is the least efficient, with a cost-to-revenue variance of -47.20%. Conversely, the UE00 & UW00 are significantly leaner, maintaining cost variances around -41.8%, indicating a more profitable business model despite lower overall volume.



(Fig.8)

1.3.6. What are the primary drivers of revenue leakage across Global Bike's operations, and how do discounting strategies impact the profit efficiency of its largest customers?

I created a Waterfall Chart to decompose our total revenue into three buckets: Gross Margin, Costs, and Discounts. Then, to see how this looks geographically, I built a Marimekko Chart, which is perfect for showing the proportional relationship between revenue and margin across different countries as shown in fig.9 below.



(Fig.9)

The Waterfall chart reveals the leakages out of $1.6B total revenue, costs eat up $885M, while discounts take a $48M bite. The Marimekko chart shows that DE is significantly more efficient; it contributes roughly $399M in margin compared to the US's $270M, despite the US having a massive footprint.

1.4 Critical Analysis of Using SAP Analytics Cloud

My experience working with the Global Bike dataset in SAP Analytics Cloud (SAC) was a significant step up from traditional spreadsheet analysis. While Excel is great for crunching numbers in a grid, SAC feels like a true command center for business strategy. The most impressive feature of SAC, in my experience, was the seamless integration of disparate data types. Dragging a Geo Map widget and instantly seeing the revenue distribution across the United States based on latitude and longitude was incredibly satisfying. Another highlight was the Augmented Analytics. In the time-series analysis, I didn't have to manually calculate moving averages or linear regressions. The automatic forecast handled the math, allowing me to focus on the meaning of the $398k revenue projection rather than the formula behind it. This shifts the user's role from a data processor to a data storyteller. However, the tool isn't without its hurdles. SAC requires a very disciplined approach to data modeling. If dimensions and measures aren't perfectly defined in the back end, the visualizations will break or show confusing results.

1.5 Conclusion

Applying SAP Analytics Cloud to the Global Bike dataset has demonstrated that data-driven decision-making is not just about having the right software; it is about having the right questions. The tool provides the lens, but the analyst must provide a strategic narrative to turn those numbers into meaningful organizational change. Overall, it allowed me to dig into the Global Bike data to find specific profitability leaks such as the margin-killing discounts in the US that would have remained buried in a standard spreadsheet.

Chapter 2: Business Data Analysis Using Tableau: Insights from the Sample Superstore Dataset

2.1 Tableau vs SAP Analytics Cloud

i. Data Exploration and Visualization: Tableau

Tableau is widely known for its ability to transform large datasets into clear and interactive visualizations. It allows analysts to quickly explore data through drag-and-drop functionality, making it easy to build charts, maps, and dashboards without complex coding. By visually analyzing patterns such as sales trends, regional performance, or product profitability, Tableau helps uncover insights that may not be immediately visible in raw datasets. The platform is particularly useful during the exploratory phase of analysis where analysts need to test different dimensions and measures to understand the underlying drivers of business performance.

ii. Enterprise Analytics and Storytelling: SAP Analytics Cloud

SAP Analytics Cloud (SAC) focuses on delivering enterprise-level analytics by integrating data visualization, reporting, and business planning within a single platform. While tools like Tableau are strong in exploratory visualization, SAC is designed to connect analytics with strategic decision-making processes. It enables organizations to combine operational data, financial planning, and predictive analytics into interactive dashboards and stories. Through these capabilities, SAC helps managers monitor key performance indicators, identify business opportunities, and align insights with broader organizational strategies.

2.2 Datasets and Suggested Questions

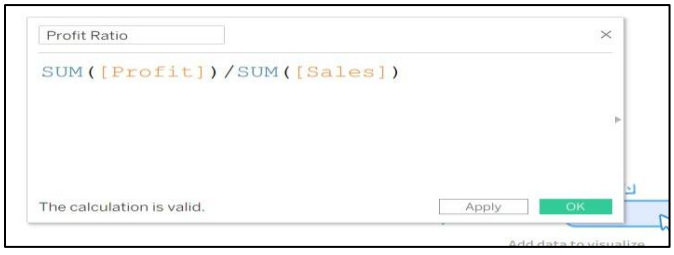
The analysis in this exercise uses the Sample Superstore dataset, which is commonly used for learning business analytics and visualization with Tableau. The dataset represents transactional sales data and contains information about orders, customers, products, locations, and financial performance. In total, the dataset includes several thousand transaction records with variables such as Order Date, Ship Date, Category, Sub-Category, Product Name, Sales, Quantity, Discount, Profit, Region, State, City, and Customer Segment. These variables allow the data to be analyzed from multiple perspectives including time, geography, product categories, and financial performance metrices. Within the dataset, Sales, Profit, Quantity, and Discount act as the key measures because they represent numerical values that can be aggregated and analyzed. Meanwhile, variables such as Region, State, City, Category, Sub-Category, Segment, and Order Date function as dimensions, enabling the data to be grouped and compared across different business contexts. The dataset's purpose is to help analysts identify patterns in sales performance, profitability, and product trends to support better business decision-making. However, the dataset also has certain limitations. Since it is a sample dataset, it does not include additional operational variables such as inventory levels, marketing expenditures, or customer demographics that could provide deeper insights into the drivers of profitability. Furthermore, the data represents a simplified business environment, meaning that real-world retail datasets would typically include larger volumes of data and more complex relationships. Despite these limitations, the dataset provides a strong foundation for exploring sales performance, profitability patterns, and regional differences using Tableau visualizations.

- Which Geographic Regions Generate the Highest Profit Margin and which state had highest profit margin in the east market?
- How do sales and profit vary across different customer segments over years?
- Which state contributes the most to sales?
- Does faster shipping lead to higher sales or profitability?
- What is the Relationship Between Discount and Profit?
- Which product sub-categories act as the primary profit drivers for the business?

2.3 Deriving Business Insights and Findings

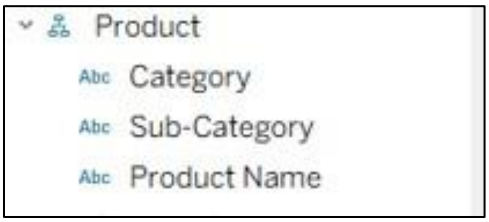
2.3.1. Which Geographic Regions Generate the Highest Profit Margin and which state had highest profit margin in the east market?

Before the map could work, I had to build the underlying logic, so I created a Calculated Field named "Profit Ratio" as shown in fig. 9. It is vital to use SUM here rather than just (Profit)/(Sales) to ensure that Tableau aggregates the data correctly when we look at the state level.



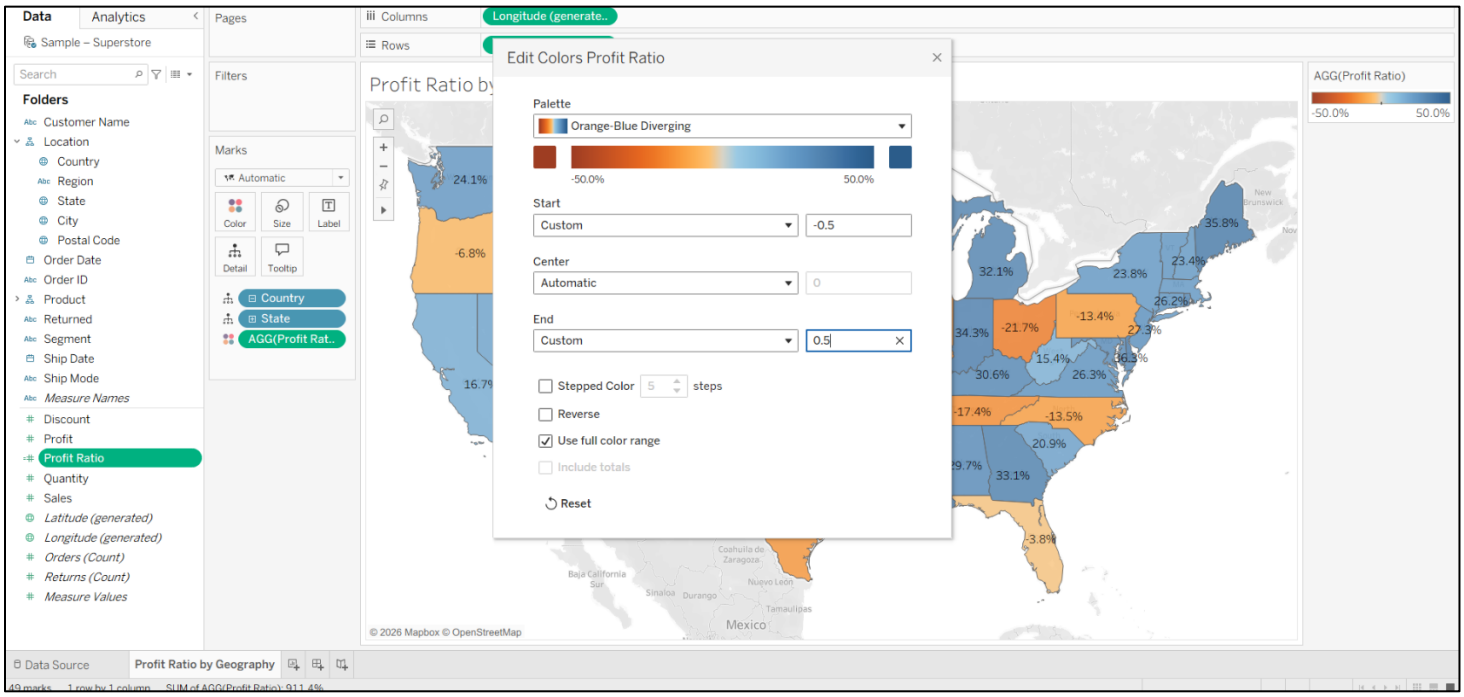
(Fig.9)

Then I created a Hierarchy in which I created a Product Hierarchy. By dragging sub-categories (like Category, Sub-Category and Product Names) under the main "Product" category, I built a "drill-down" feature as shown in fig.10



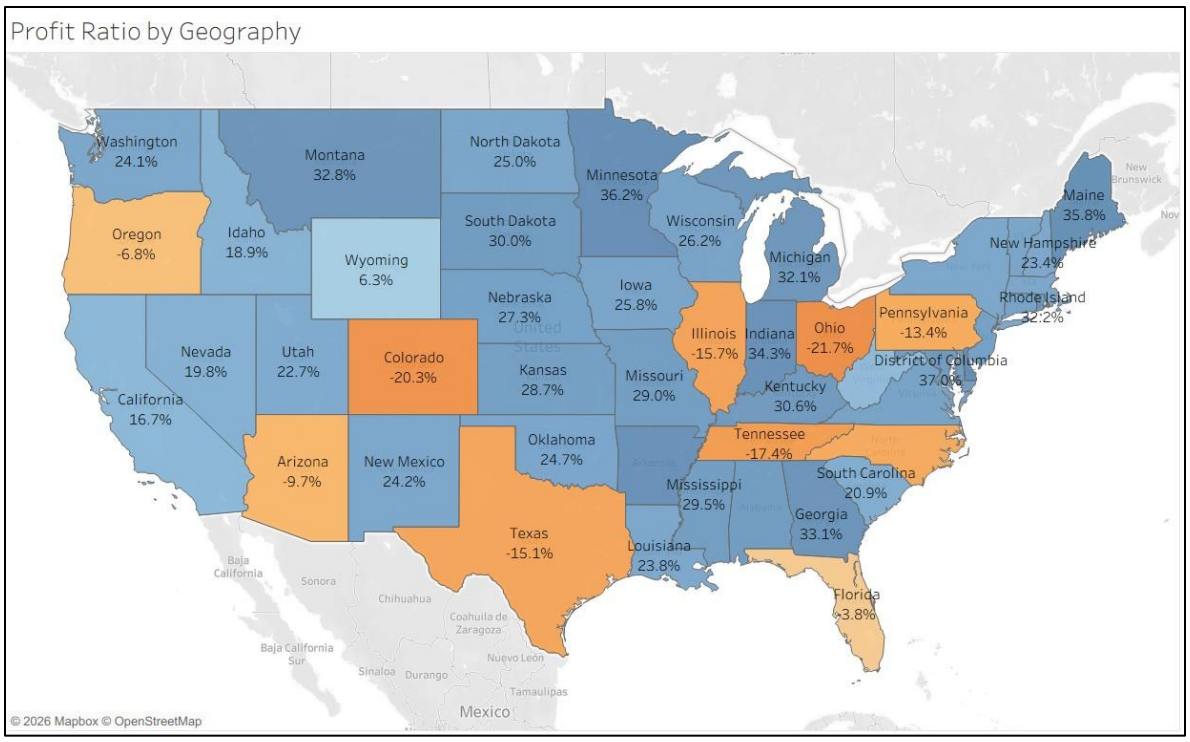
(Fig.10)

To create the "Profit Ratio by Geography" map, I started by double-clicking the State dimension, which automatically generated the map view using Tableau's built-in geographic coordinates. I then dragged my custom Profit Ratio calculated field onto the Color mark. I then used a Diverging Palette (Orange-Blue). This is a visual best practice because it clearly separates "danger zones" (negative profit in orange) from "success zones" (positive profit in blue) as shown I dragged both State and Profit Ratio onto the Label mark, so the exact percentages are visible at a glance as shown in fig.11 below.



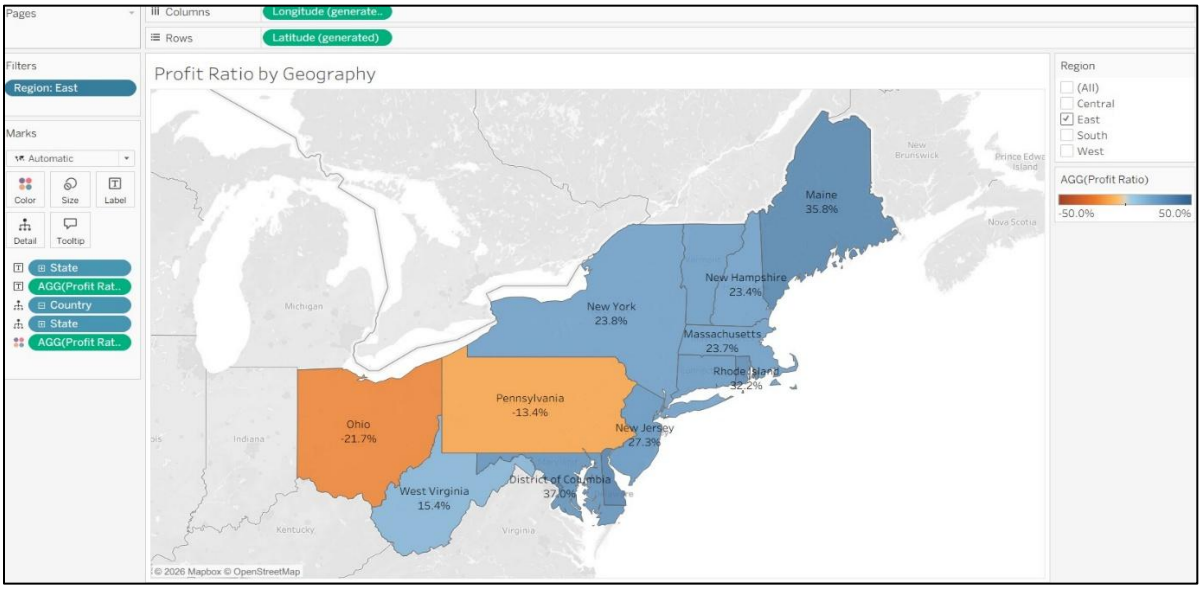
(Fig.11)

As shown in fig.12, the East Coast shows strong performance, particularly in states like Maine (35.8%) and the District of Columbia (37.0%). In the "Orange" Danger Zones, there is a significant profitability drain in the Midwest and South. Ohio (-21.7%) and Pennsylvania (-13.4%) are deep in the orange. This suggests that while sales might be happening there, the cost of doing business or the heavy discounting in those states is killing the bottom line. Interestingly, right next to struggling Ohio, Indiana is thriving with a 34.3% profit ratio. This indicates that the issue isn't regional economy-wide, but likely specific within those individual states.



(Fig.12)

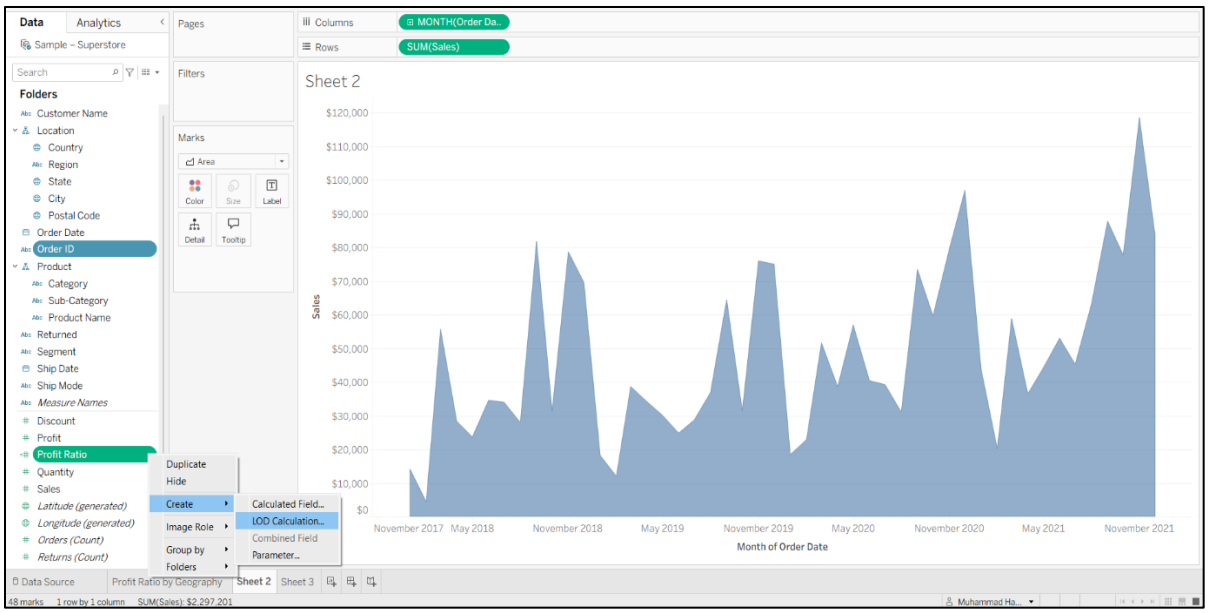
I added a Region Filter to the right-hand side, allowing to zoom into specific areas like the "East," which cleans up the view and focuses the analysis on local performance as shown in fig.13 below.



(Fig.13)

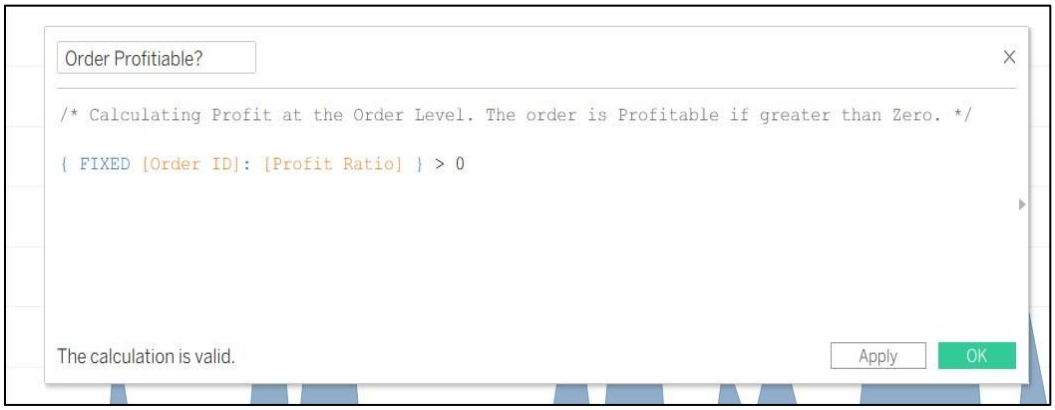
2.3.2. How Do Sales and Profit Vary Across Different Customer Segments over years?

Before building the final visualization, I created a Level of Detail (LOD) expression called "Order Profitable?". Using a FIXED LOD calculation, I instructed Tableau to evaluate profitability specifically at the Order ID level, rather than just looking at individual line items. The formula used can be seen in Figs. 14 and 15.



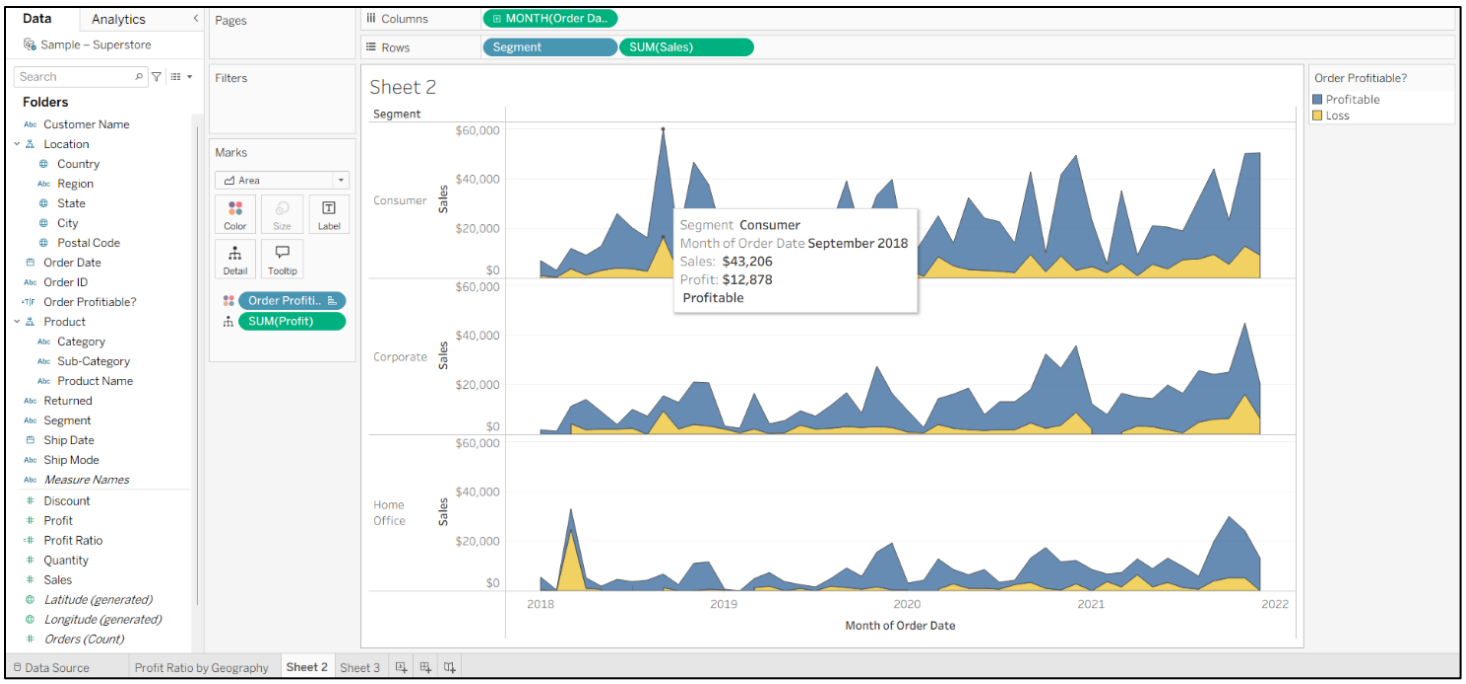
(Fig.14)

This expression creates a boolean dimension that categorizes every unique order as either a "Profitable" order (True) or a "Loss" (False). This step is crucial for the rubric as it demonstrates advanced technical competency in Tableau.



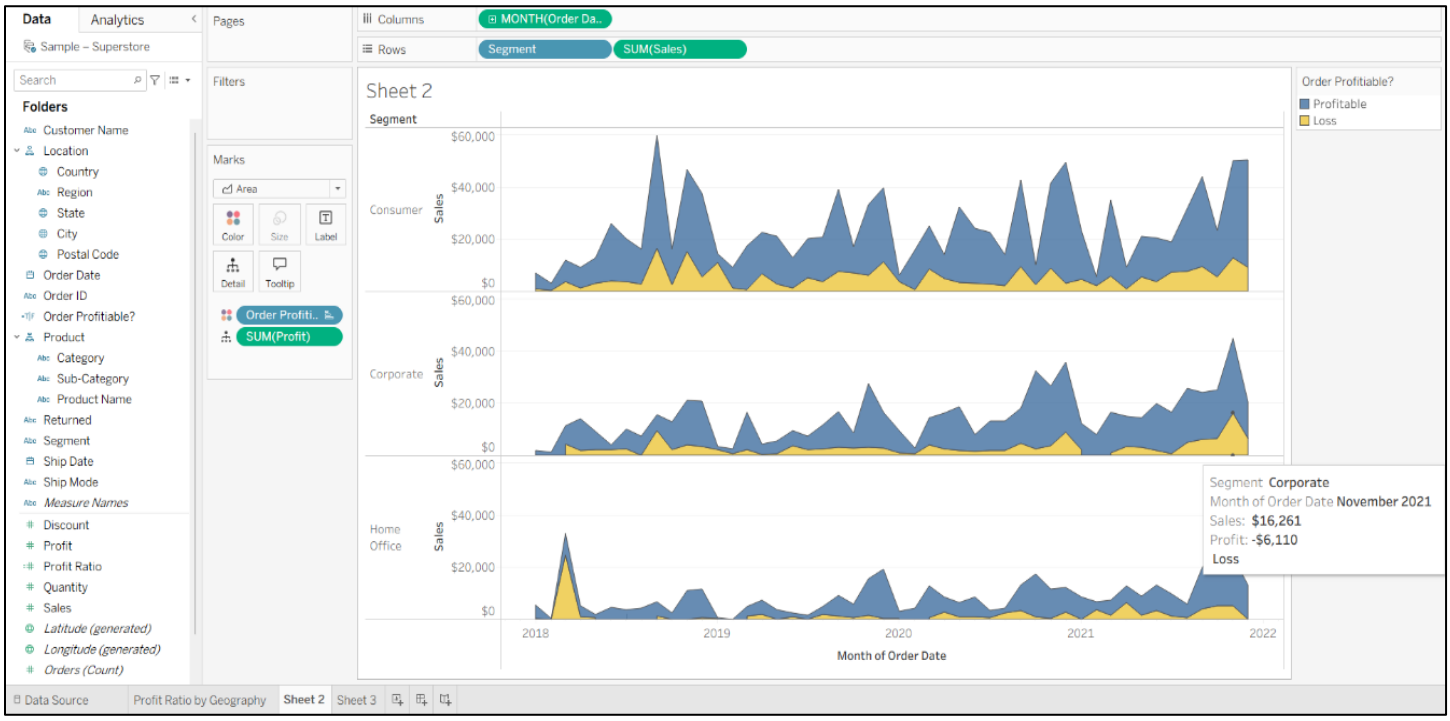
(Fig.15)

With the LOD expression ready, I constructed an Area Chart to visualize the relationship between sales and profitability over several years. I placed Month of Order Date on the Columns shelf to show a continuous timeline from 2018 through late 2021 and used SUM(Sales) on the Rows shelf to track total revenue volume. I then dragged the Segment dimension into the Rows shelf to create three distinct segments for the Consumer, Corporate, and Home Office categories. I placed the newly created “Order Profitable? – LOD expression” onto the Color mark. This instantly split each area chart into two colors, blue representing sales from profitable orders and yellow/gold representing sales that resulted in a loss as shown in fig. 16 below.



(Fig.16)

The Consumer segment is the largest driver of volume, showing significant seasonal peaks. For example, in September 2018, this segment hit a peak of $43,206 in sales with a solid profit of $12,878. While the Corporate segment has high sales potential, it also shows significant pockets of loss.



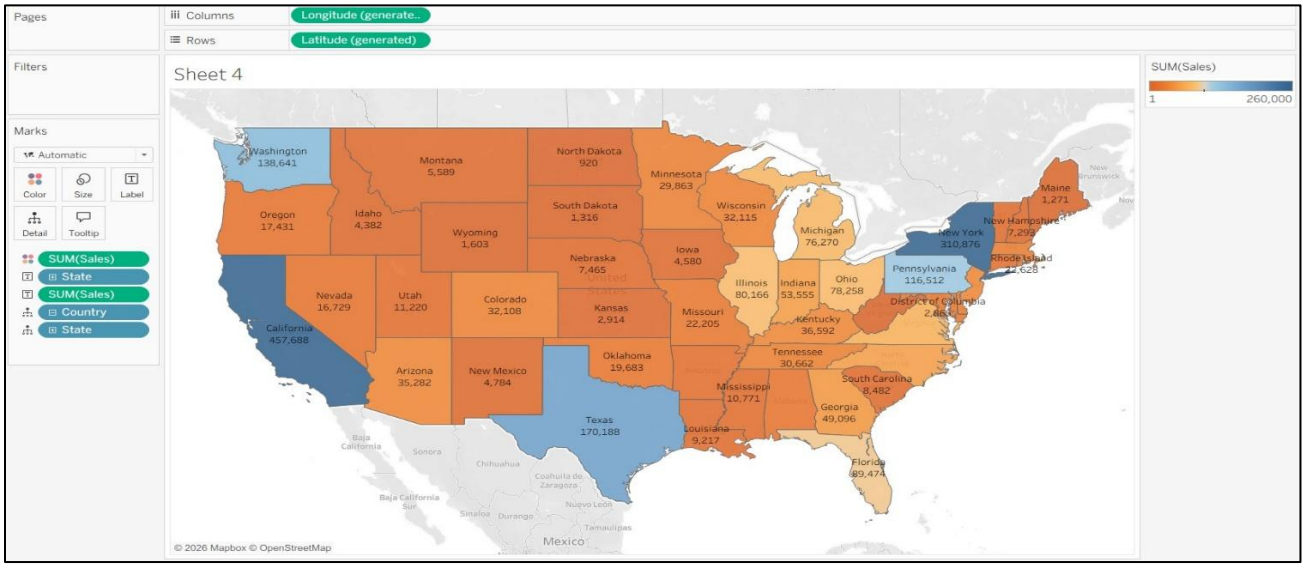
(Fig.17)

As shown in fig.17, in November 2021, despite generating $16,261 in sales, the segment recorded a net loss of $6,110. This indicates that high sales in this segment do not always guarantee a healthy bottom line, likely due to heavy bulk discounting. The Home Office segment is the smallest in terms of raw sales volume but appears to have the most consistent ratio of blue-to-yellow, suggesting more stable, higher-margin transactions with fewer instances of extreme loss compared to the larger

segments. Across all three segments, there is a visible upward trend in sales volume moving into late 2021, but the yellow area (losses) also expands during these high-volume periods, suggesting that aggressive end-of-year sales tactics may be eroding profit margins.

2.3.3. Which state contributes the most to sales?

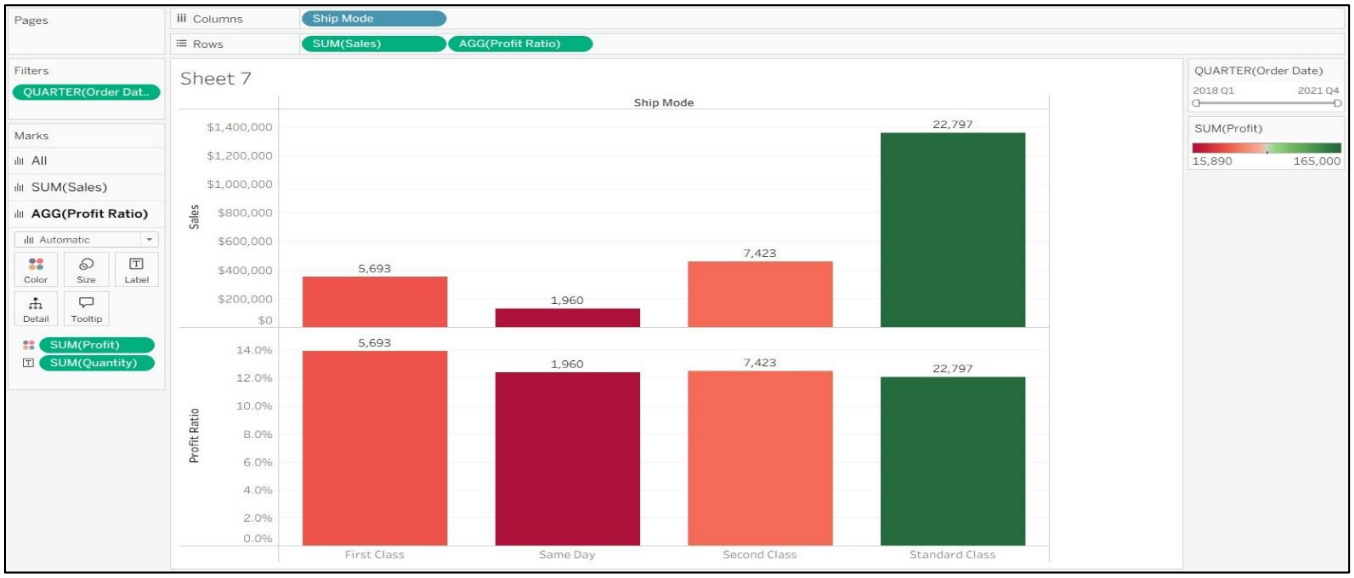
To determine the geographic distribution of revenue, a map visualization was created to show total sales volume across the United States. California is the largest contributor to sales with a total of $457,688, followed by New York at $310,876. Texas ($170,188) and Washington ($138,641) also show high sales volumes, represented by the darker blue shades on the map. Lower Volume States such as North Dakota ($920) and West Virginia ($1,210) contribute the least to the overall sales figures as shown in Fig. 18 below.



(Fig.18)

2.3.4. Does faster shipping lead to higher sales or profitability?

To determine if faster shipping leads to higher sales or profitability, I analyzed the relationship between Ship Mode, Sales, and Profit Ratio. I placed Ship Mode on the Columns shelf and added SUM(Sales) and AVG(Profit Ratio) to Rows as shown in fig.19 below.

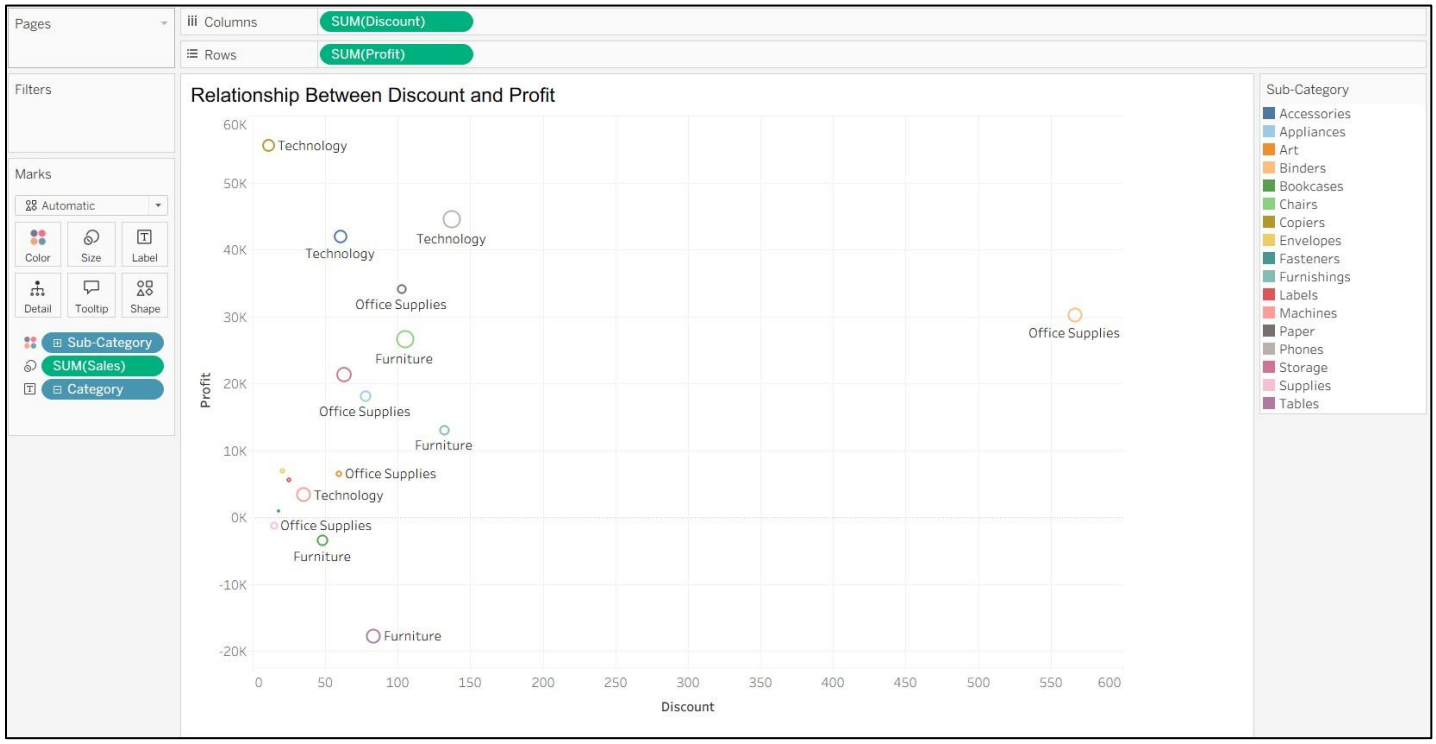


(Fig.19)

Faster shipping does not automatically lead to higher sales. Standard Class shipping accounts for the vast majority of revenue ($1.36M), while Same Day shipping contributes the least ($128K). This suggests customers are generally willing to wait longer for a lower shipping cost. As shown in fig. 19, the most profitable shipping tier is actually Second Class, boasting a 16.1% profit ratio. Interestingly, Same Day shipping (12.4%) is only marginally more profitable than Standard Class (12.0%), likely because the high operational costs and logistics premiums required for rapid delivery eat into the margins. For Global Bike Inc., Second Class shipping represents the "Goldilocks" zone as it offers a balance of speed that customers value enough to maintain healthy margins without the extreme overhead of same-day logistics.

2.3.5. What is the Relationship Between Discount and Profit?

To analyze the relationship between discounts and profitability, I utilized a scatter plot to visualize how varying discount levels impact the overall profit across different product categories. To see the distribution clearly, I dragged Category for Color, allowing me to distinguish between furniture, office supplies, and technology as shown in fig.20 below.

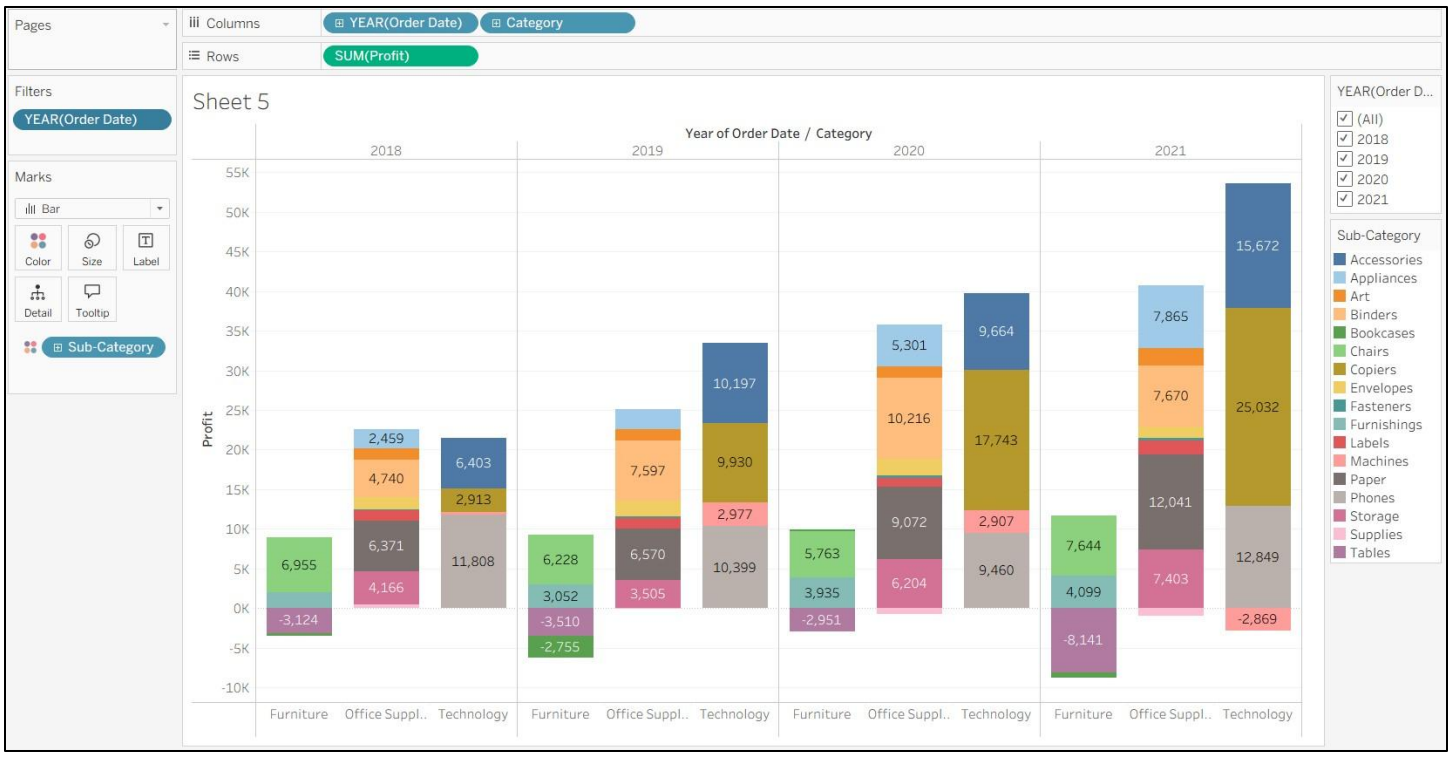


(Fig.20)

Most products sold with a discount of 0% to 20% remain in the "profitable" zone (above the $0 line). However, once discounts exceed 30%, almost all transactions result in a net loss. Technology and Office Supplies tend to maintain higher profitability even at low discount levels. Furniture appears highly sensitive to discounting; large clusters of furniture items (represented in a specific color) fall deep into negative profit as soon as aggressive discounts are applied. The most significant losses (the lowest points on the Y-axis) are directly associated with the highest discount levels (70%–80%).

2.3.6. Which product sub-categories act as the primary profit drivers for the business?

To make this visual, I placed Year of Order Date and Category on the on the Columns and then dragged Profit to the Rows. To make the distinction clear, I put sub-category in the color mark as shown in fig.21 below.



(Fig.21)

While Copiers might not always have the highest daily sales volume compared to Phones, they maintain a remarkably high and stable Profit Ratio (often peaking above 40%). This makes them the most profitable sub-category over the long term. Phones generate massive revenue spikes (the tall bars), but their Profit Ratio line sits lower than Copiers. This suggests that while Phones bring in the most cash, they operate on thinner margins. The Accessories sub-category shows very consistent growth and a healthy, stable profit ratio.

2.4 Critical Analysis: Tableau vs. SAP Analytics Cloud

Tableau is very effective for exploratory data analysis and visualization. Its drag-and-drop interface makes it easy to create charts, maps, and dashboards, which helps analysts quickly identify patterns such as regional differences in sales or profitability. It is especially useful for interactive exploration, filtering data, and presenting insights in a clear visual format. However, Tableau mainly focuses on visualization and does not provide strong built-in capabilities for enterprise planning or forecasting. SAP Analytics Cloud (SAC), in contrast, is designed more for enterprise-level analytics and decision-making. It integrates visualization with planning, forecasting, and performance management, allowing organizations to connect analysis directly with business strategy. While SAC is powerful for structured reporting and planning, it can be less flexible and intuitive for quick exploratory analysis compared to Tableau. Overall, Tableau is better suited for discovering and communicating insights visually, while SAC is more appropriate for enterprise planning and strategic analytics.

2.5 Conclusion

To sum up, Tableau and SAP Analytics Cloud play different but complementary roles in the analytics process. Tableau is very effective for exploring data and creating visual dashboards that quickly reveal patterns such as regional profit differences or product performance. Its interactive visualizations make it easier to understand trends without relying on raw tables or complex calculations. SAP Analytics Cloud, however, goes beyond visualization by supporting enterprise-level

reporting and decision-making. It allows organizations to connect insights with planning and performance management. In short, Tableau helps analysts discover and communicate insights visually, while SAP Analytics Cloud helps organizations turn those insights into strategic business decisions.

Chapter 3 Reporting with SAP BW using SAP Crystal Reports

3.1 Introduction of SAP Crystal Reports

SAP Crystal Reports is a business intelligence reporting tool used to convert raw data into structured and professional reports. It allows users to connect to multiple data sources, including enterprise systems such as SAP Business Warehouse, and organize data into clear summaries and formatted reports. The tool enables users to filter data, group records, create calculations, and generate summaries such as subtotals and grand totals. These capabilities make it easier to transform large datasets into meaningful insights. In business environments, SAP Crystal Reports is commonly used for sales reporting, financial analysis, and operational performance monitoring, helping managers interpret data and support informed decision-making.

3.2 Dataset and Questions

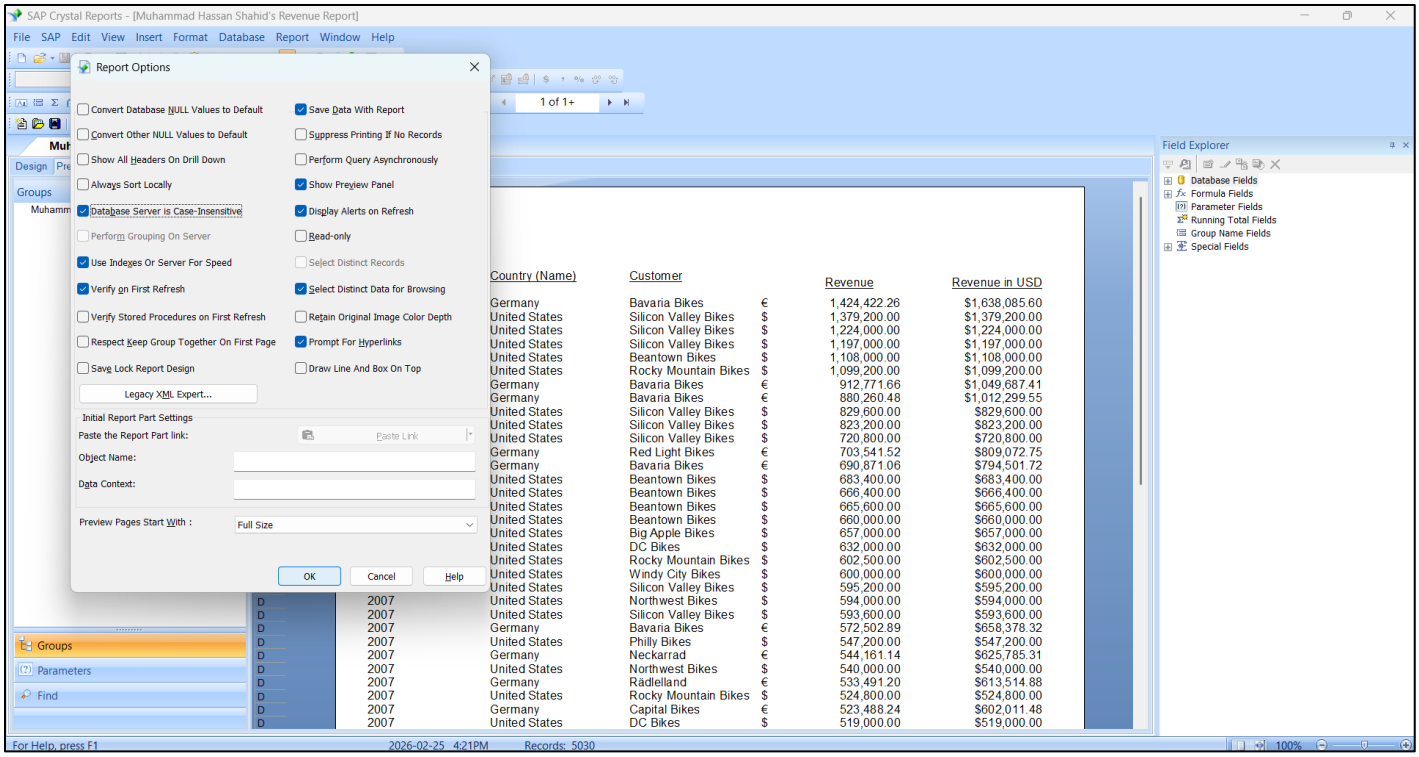
The dataset used in this exercise comes from the SAP Business Warehouse (BW) Composite Provider "MU0 Sales Data Composite Provider," which contains sales transaction data for Global Bike Inc. The data is designed to support reporting and performance analysis for the company's sales team. The dataset includes several dimensions and measures. The main dimensions are Calendar Year, Country, Customer, and Division. The primary measure (key figure) is Sales Revenue, representing the total sales generated by customers across different countries and years. However, the dataset has some limitations. It does not include additional variables such as product categories, store-level profitability, or detailed customer segmentation, which could provide deeper business insights if included in future analyses. Based on the reporting requirements, the following questions guide the report design:

- How can record selection be applied in SAP Crystal Reports to filter the dataset for a specific division?
- How can the sales data be organized and grouped by Calendar Year and Country in the report?
- How can a clear and professional report header be added to improve report readability?
- How can a page footer with page numbers and copyright information be included in the report?

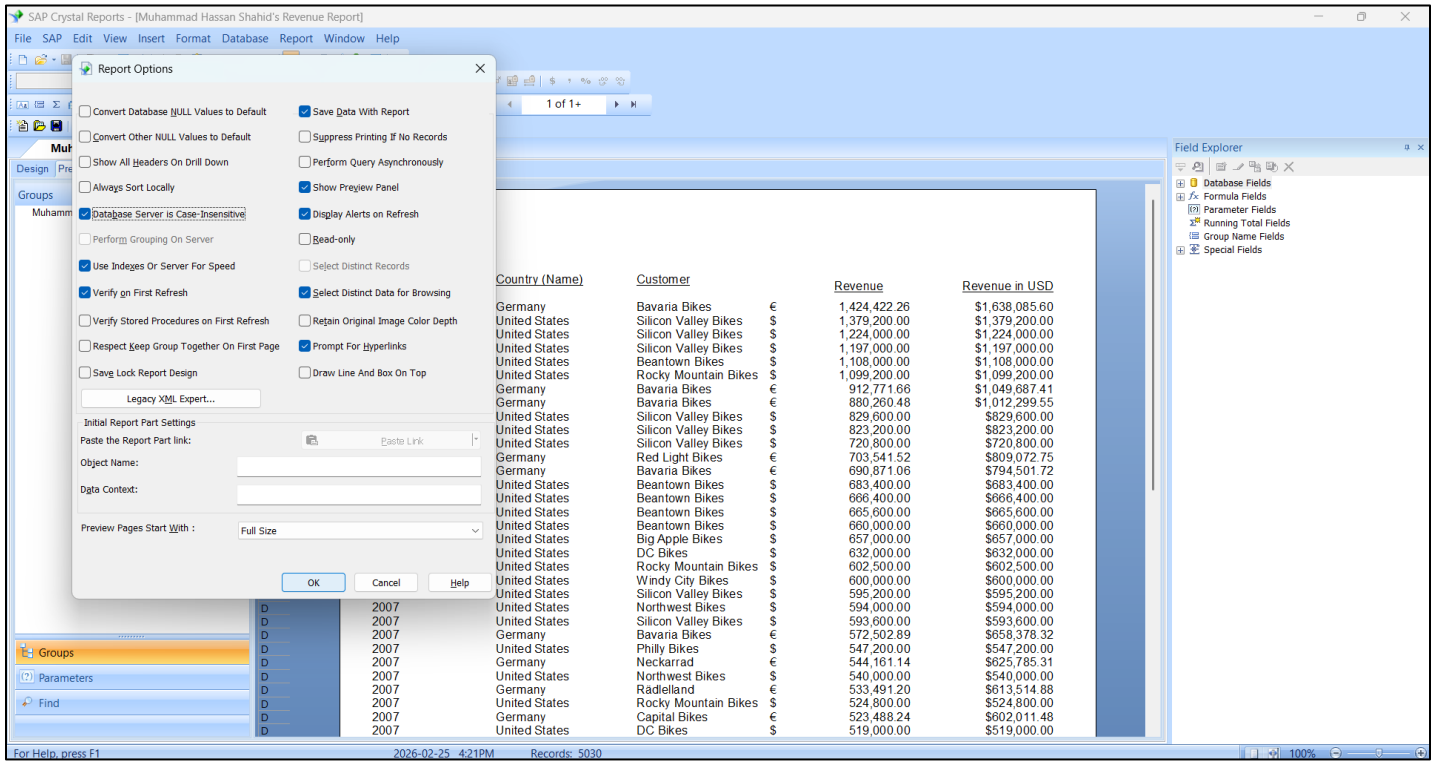
3.3 Application of Report Designing and Formatting in SAP Crystal Reports

3.2.1. How can record selection be applied in SAP Crystal Reports to filter the dataset for a specific division?

I first navigated to File → Report Options. In the options dialog, I enabled the checkbox for Database Server is Case-Insensitive and clicked "Ok" as shown in figs 22 and 23.

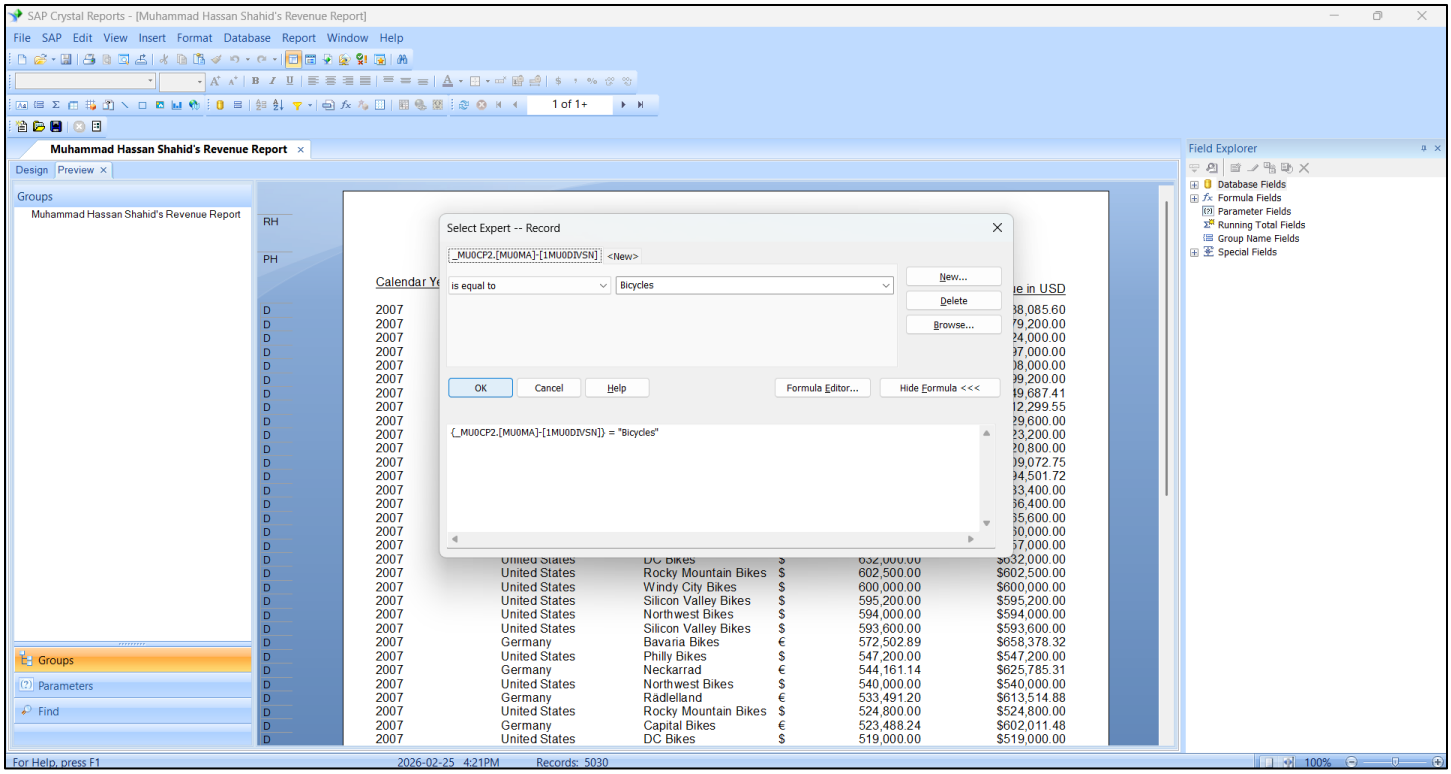


(Fig.23)



(Fig.24)

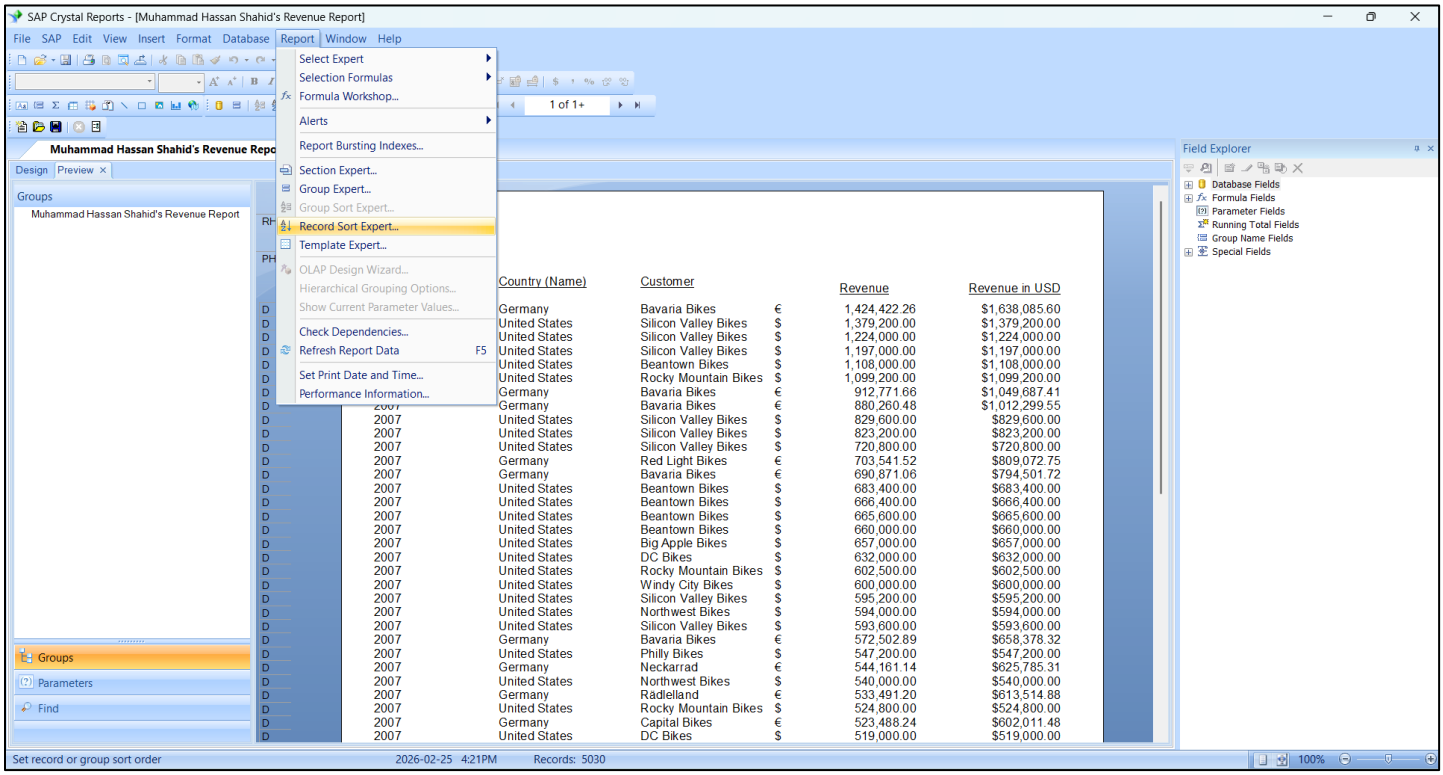
Then I opened the Select Expert (Report → Select Expert → Record) and created a New tab. From the field list, I navigated through the MU0 Product table to select the Division (Name) field. In the criteria dropdown, I chose is equal to and then selected Bicycles from the secondary dropdown list. After clicking OK, I switched to the Preview Tab. When prompted, I chose Refresh Data to force Crystal Reports to re-query the data warehouse.



(Fig.25)

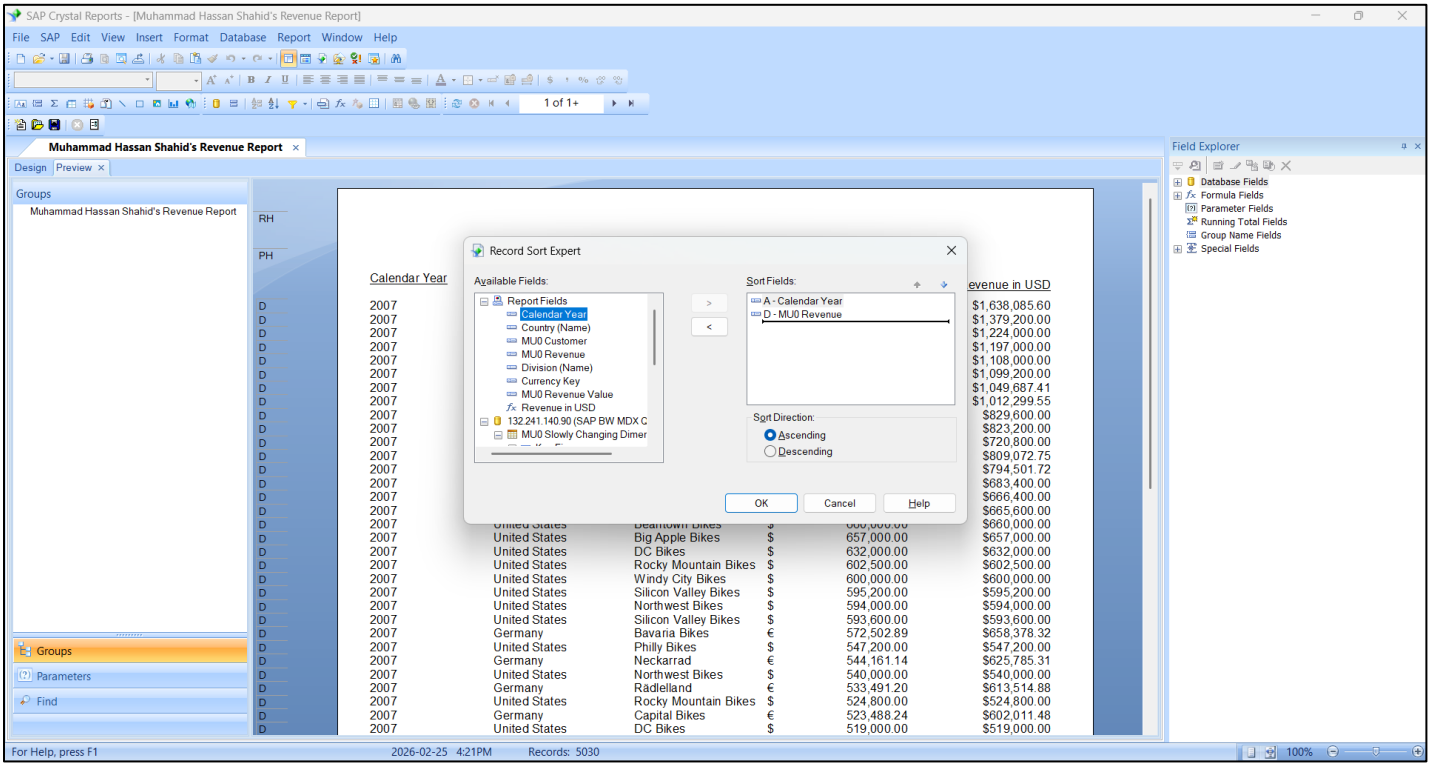
3.3.2. How can the sales data be organized and grouped by Calendar Year and Country in the report?

I navigated to the top menu and selected Report → Record Sort Expert. Fig.26 shows the exact moment I accessed the Record Sort Expert from the main Report menu. In the Record Sort Expert dialog box, I located the Calendar Year field under "Available Fields" and moved it to the "Sort Fields" list using the arrow button. I set the Sort Direction to Ascending, ensuring the report starts with the earliest year.



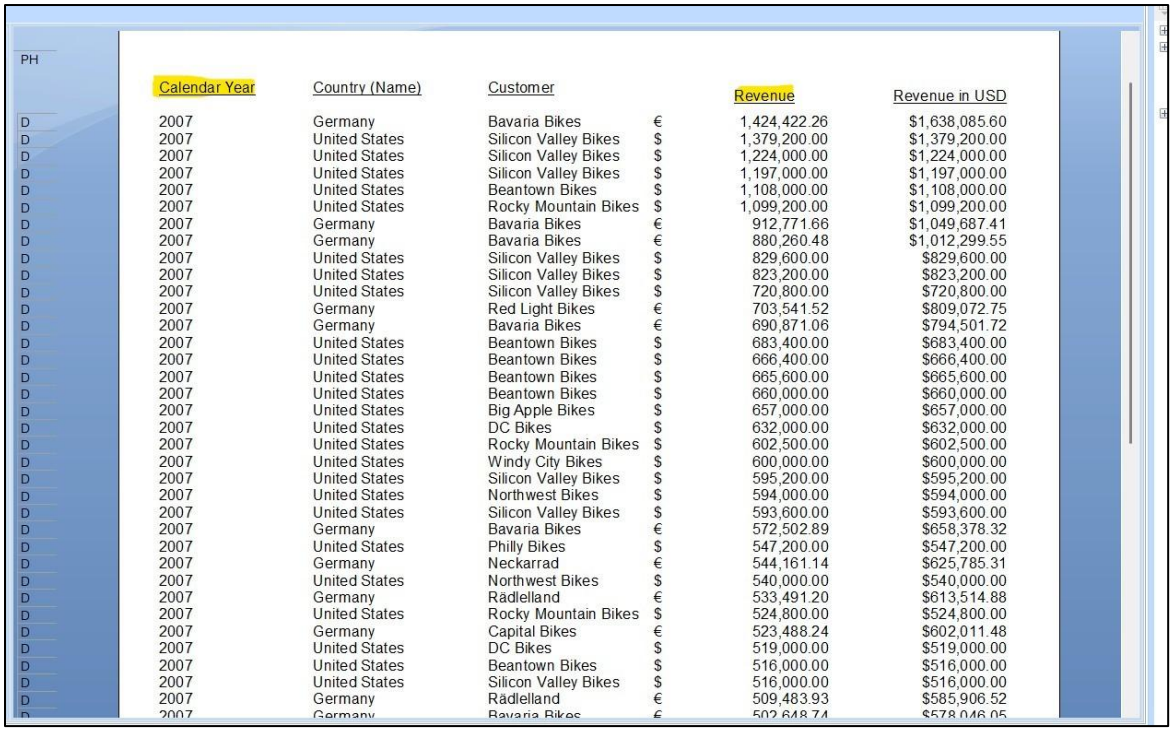
(Fig.26)

Fig.27 is the most important "technical" visual; it confirms the logic I applied; sorting first by Year (A) and then by Revenue (D). I then added the MU0 Revenue field to the sort list. Crucially, I changed its Sort Direction to Descending. This ensures that within each calendar year, the highest-grossing transactions appear at the top of the list.



(Fig.27)

Once I clicked OK, the report layout updated immediately. As seen in Figure 28, the data is now perfectly organized: all records are grouped under 2007 (Ascending), but the Revenue column shows values decreasing from the highest amount ($1,424,422.26) downward (Descending).

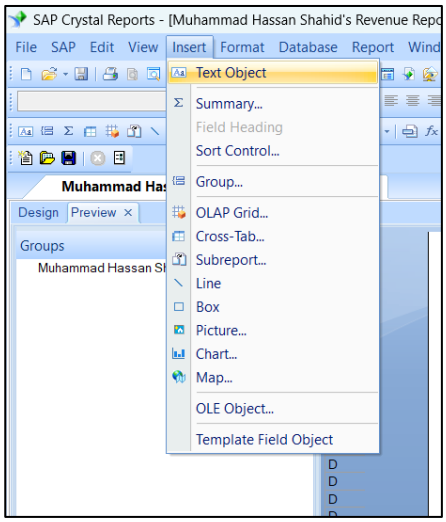


Calendar Year	Country (Name)	Customer		Revenue	Revenue in USD
2007	Germany	Bavaria Bikes	€	1,424,422.26	$1,638,085.60
2007	United States	Silicon Valley Bikes	$	1,379,200.00	$1,379,200.00
2007	United States	Silicon Valley Bikes	$	1,224,000.00	$1,224,000.00
2007	United States	Silicon Valley Bikes	$	1,197,000.00	$1,197,000.00
2007	United States	Beantown Bikes	$	1,108,000.00	$1,108,000.00
2007	United States	Rocky Mountain Bikes	$	1,099,200.00	$1,099,200.00
2007	Germany	Bavaria Bikes	€	912,771.66	$1,049,687.41
2007	Germany	Bavaria Bikes	€	880,260.48	$1,012,299.55
2007	United States	Silicon Valley Bikes	$	829,600.00	$829,600.00
2007	United States	Silicon Valley Bikes	$	823,200.00	$823,200.00
2007	United States	Silicon Valley Bikes	$	720,800.00	$720,800.00
2007	Germany	Red Light Bikes	€	703,541.52	$809,072.75
2007	Germany	Bavaria Bikes	€	690,871.06	$794,501.72
2007	United States	Beantown Bikes	$	683,400.00	$683,400.00
2007	United States	Beantown Bikes	$	666,400.00	$666,400.00
2007	United States	Beantown Bikes	$	665,600.00	$665,600.00
2007	United States	Beantown Bikes	$	660,000.00	$660,000.00
2007	United States	Big Apple Bikes	$	657,000.00	$657,000.00
2007	United States	DC Bikes	$	632,000.00	$632,000.00
2007	United States	Rocky Mountain Bikes	$	602,500.00	$602,500.00
2007	United States	Windy City Bikes	$	600,000.00	$600,000.00
2007	United States	Silicon Valley Bikes	$	595,200.00	$595,200.00
2007	United States	Northwest Bikes	$	594,000.00	$594,000.00
2007	United States	Silicon Valley Bikes	$	593,600.00	$593,600.00
2007	Germany	Bavaria Bikes	€	572,502.89	$658,378.32
2007	United States	Philly Bikes	$	547,200.00	$547,200.00
2007	Germany	Neckarrad	€	544,161.14	$625,785.31
2007	United States	Northwest Bikes	$	540,000.00	$540,000.00
2007	Germany	Rädlelland	€	533,491.20	$613,514.88
2007	United States	Rocky Mountain Bikes	$	524,800.00	$524,800.00
2007	Germany	Capital Bikes	€	523,488.24	$602,011.48
2007	United States	DC Bikes	$	519,000.00	$519,000.00
2007	United States	Beantown Bikes	$	516,000.00	$516,000.00
2007	United States	Silicon Valley Bikes	$	516,000.00	$516,000.00
2007	Germany	Rädlelland	€	509,483.93	$585,906.52
2007	Germany	Bavaria Bikes	€	502,648.74	$578,046.05

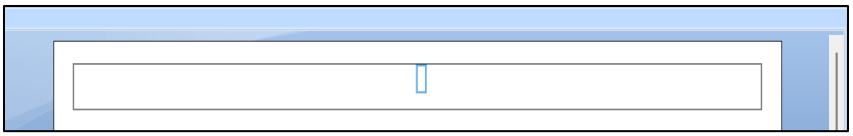
(Fig.28)

3.3.3. How can a clear and professional report header be added to improve report readability?

As shown in fig.29, I navigated to Insert → Text Object and placed the new object in the center of the Report Header (RH) section. I entered the title "Hassan's Report" into the text box. To ensure it looked like a formal header, I used the sizing handles to stretch the text box across the entire width of the report header section as shown in fig.30. Using the alignment icons in the toolbar, I centered the text within the object to maintain visual balance.

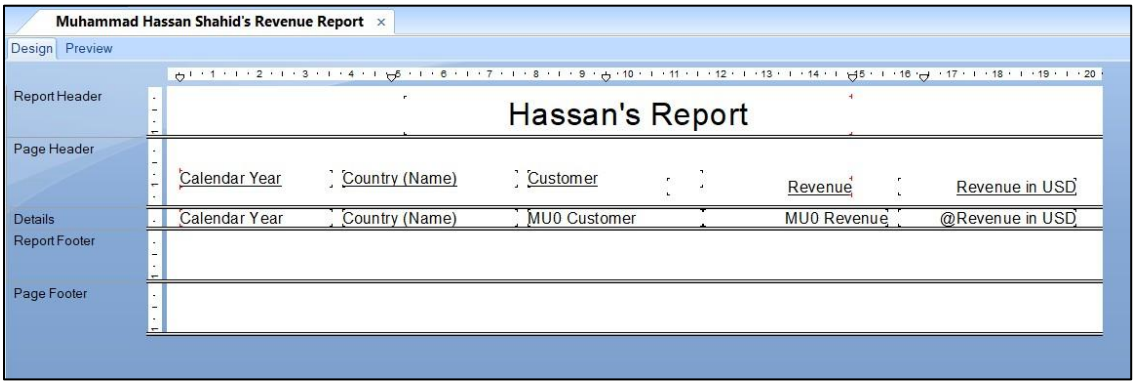


(Fig.29)



(Fig.30)

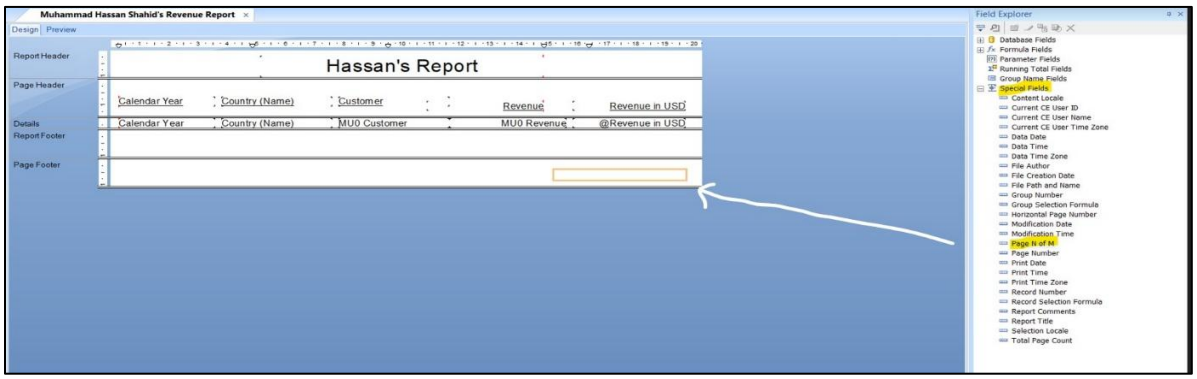
I highlighted the text, accessed Text Formatting, and increased the font size to 20pt to make the title prominent as shown in fig.31 below.



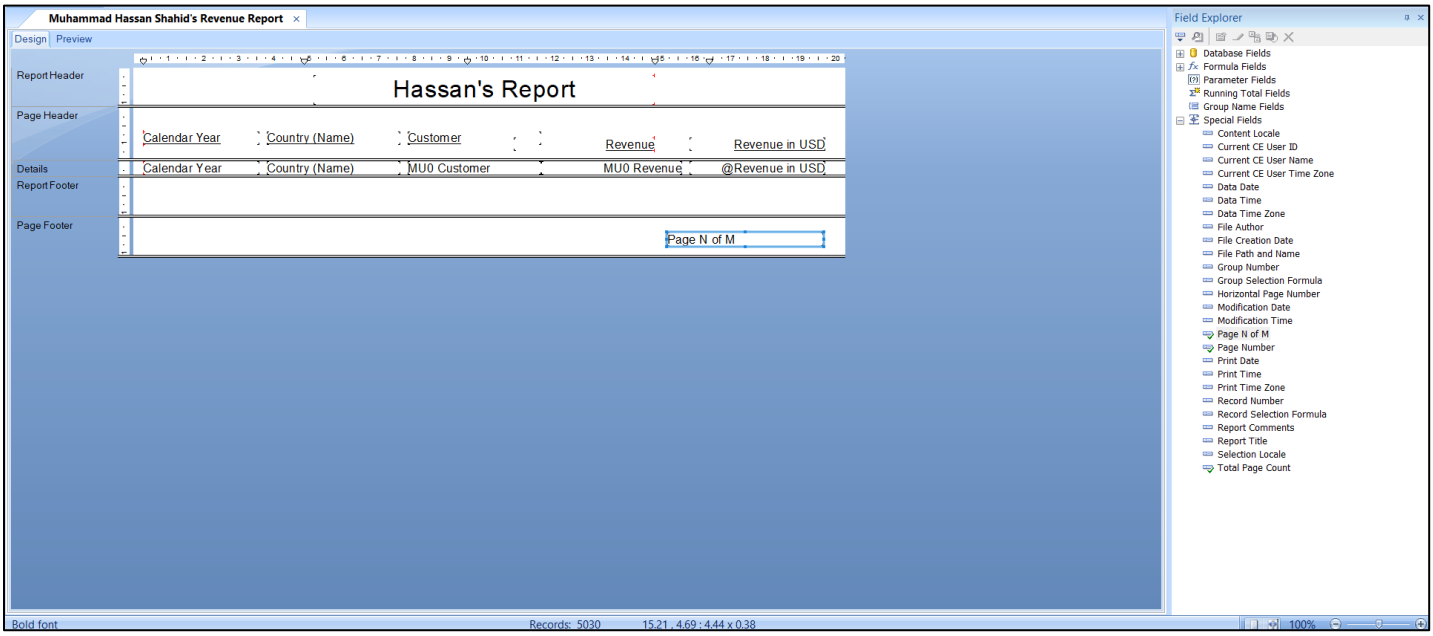
(Fig.31)

3.3.4. How can a page footer with page numbers and copyright information be included in the report?

In the Field Explorer, I navigated to the Special Fields node. I selected the Page N of M field, which automatically calculates current and total pages and dragged it into the right-hand side of the Page Footer (PF) section as shown in figs. 32 and 33.

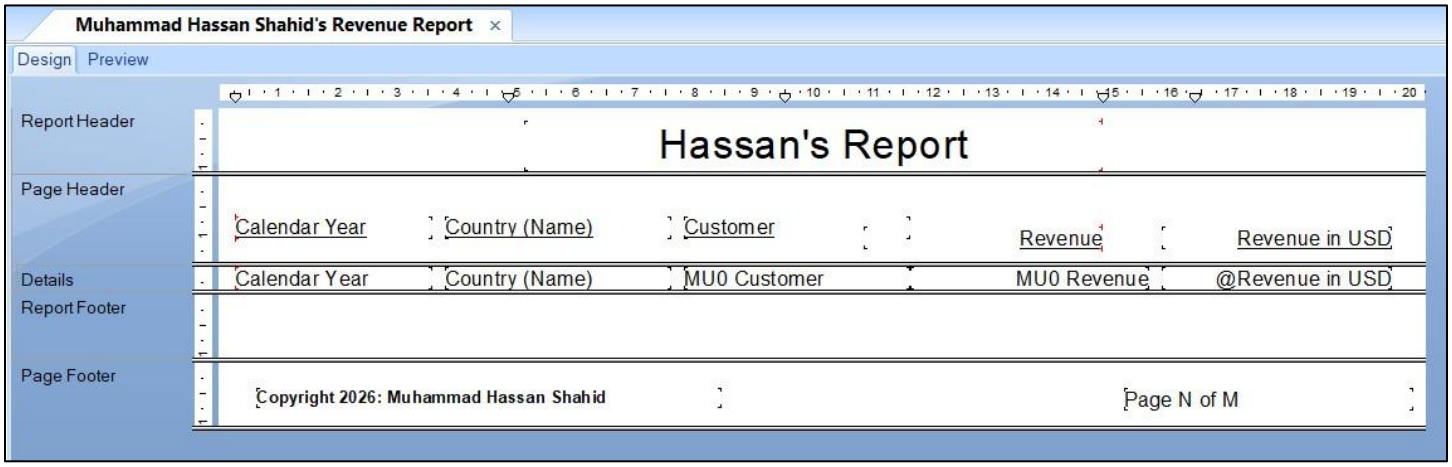


(Fig.32)



(Fig.33)

I then inserted a new Text Object into the center of the Page Footer. I entered the text "Copyright 2026: Muhammad Hassan Shahid," centered the object using the drag handles, and applied bold to match the styling of the page numbers. I also toggled between the Design and Preview panes to verify that the footer objects were correctly aligned and visible across different pages before saving the final report as shown in fig. 34.



(Fig.34)

3.4 Critical Analysis: SAP Crystal Reports

SAP Crystal Reports is primarily designed for structured business reporting rather than exploratory data analysis. It allows report designers to connect directly to enterprise data sources such as SAP Business Warehouse and generate highly formatted reports with grouping, sorting, and summarized calculations. One of the strengths of SAP Crystal Reports is its ability to produce professional, well-formatted reports that follow organizational reporting standards. However, compared to modern visualization tools, Crystal Reports can feel less interactive and more rigid when exploring data. It is most effective for generating standardized operational reports, but it is less suitable for quick exploratory analysis or interactive dashboards.

3.5 Conclusion

Overall, SAP Crystal Reports is a powerful tool for creating structured and professional business reports. In this exercise, it helped transform raw sales data from the SAP Business Warehouse into a well-organized report that highlights revenue performance across years and countries. Features such as grouping, sorting, and summarized totals made it easier to interpret the data and present it in a clear reporting format. While the tool may not provide the same level of interactivity as modern visualization platforms, it remains highly valuable for producing reliable and standardized enterprise reports used by management for monitoring business performance.